Safety-Critical Control with Control Barrier Functions: *Backup Construction* and **Robustness**

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Safety-critical systems require ***formal guarantees*** in **control design.**

Contents

- Formally guaranteeing safety with ***Control Barrier Functions***.
- Certifying CBF with controlled invariant set (***backup constructed***).
- Robustfying the scheme against disturbances using ***Input-to-state safety.***
- Safe car following example.

Formal Safety

System $\dot{x} = f(x),$

with an initial set $x_0 \in \mathcal{X}_0 \subset \mathbb{R}^n$

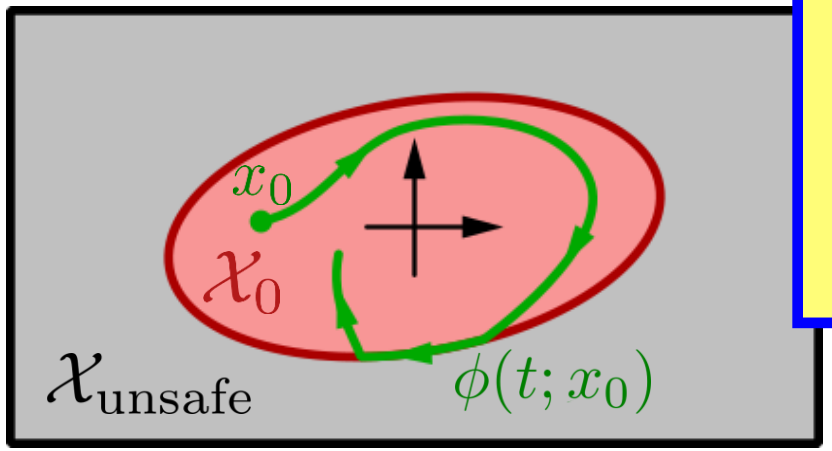
and an unsafe set $\mathcal{X}_{\text{unsafe}} \subset \mathbb{R}^n$

such that $\mathcal{X}_0 \cap \mathcal{X}_{\text{unsafe}} = \emptyset$

→ Solution: $\phi(t; x_0)$

Safety

$$\forall x_0 \in \mathcal{X}_0 \;\Rightarrow\; \phi(t; x_0) \notin \mathcal{X}_{\text{unsafe}}, \;\forall t \geq 0.$$



Function-based definition:

Let $S = \{x \mid h(x) \geq 0\}$ with $h : \mathbb{R}^n \to \mathbb{R}, \; S \subseteq \mathcal{X}_0.$

System is safe if

$$\forall x_0 \in S, \; h(\phi(t; x_0)) \geq 0, \; \forall t \geq 0.$$

Set-Theoretic Safety Verification

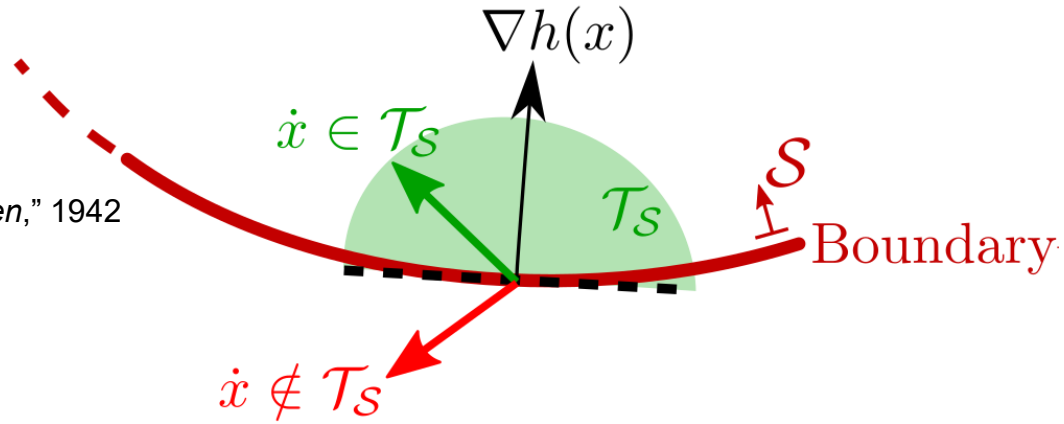
*"…study properties of the system without the need to compute the trajectory explicitly."**

*S. Prajna, et al., "*A framework for worst-case and stochastic safety verification using barrier certificates*." 2007.

1942 ⇩

➢ Nagumo's theorem

M. Nagumo, "*Über die lage der integralkurven gewöhnlicher differentialgleichungen*," 1942



Definition: *Tangent cone*

Given a set S, the tangent cone to S at $x \in \mathbb{R}^n$:

$$\mathcal{T}_S(x) \triangleq \left\{ z \;\middle|\; \liminf_{\delta \to 0} \frac{\operatorname{dist}(x + \delta z, S)}{\delta} = 0 \right\}$$

If h is Lipschitz OR smooth

$$\mathcal{T}_S(x) \subseteq \left\{ v \;\middle|\; h^\circ(x; v) \geq 0 \text{ OR } \nabla h(x)^T v \geq 0 \right\}.$$

Theorem: Assume $\phi(t; x_0)$ is unique $\forall x_0 \in \mathbb{R}^n$ and $t \geq 0$.
System is safe w.r.t. S if and only if

$$\dot{x} = f(x) \in \mathcal{T}_S(x), \quad \forall x \in S.$$

If non-degenerate boundary
$(0 \notin \partial^\circ h(x) \text{ OR } \nabla h(x) \neq 0)$

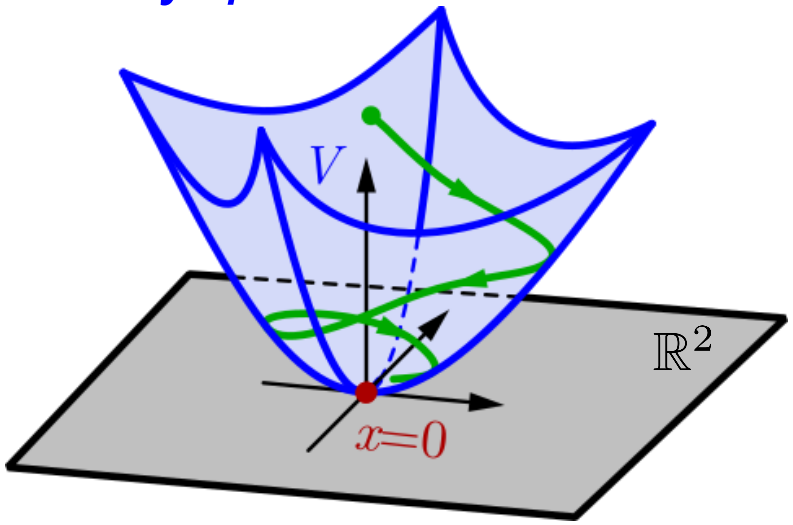
F. Blanchini and S. Miani, *Set-Theoretic Methods in Control*, 2nd ed. Cham, Switzerland: Springer, 2015

Control Barrier Functions: Key idea inspired from *Lyapunov Functions*

Control system: $\dot{x} = f(x, u), \quad u \in \mathbb{U} \subseteq \mathbb{R}^m.$

Goal: Find $u = k(x) \implies f(x, k(x))$ is safe.

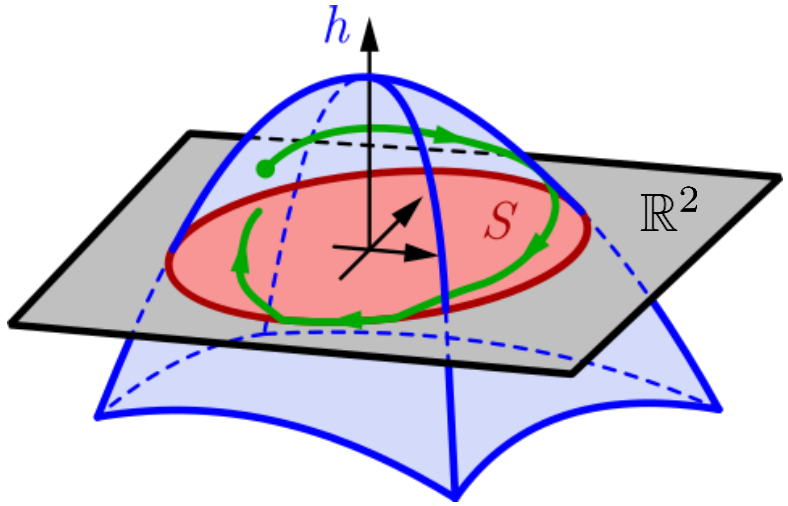
Lyapunov Functions



$$V \in C^1, \; V(0) = 0, \; V(x) > 0, \forall x \in \mathbb{R}^n$$
$$\nabla V(x) \cdot \dot{x} \leq -\lambda V(x), \quad \forall x \in \mathbb{R}^n$$

Approach equilibrium exponentially with a *minimum rate*.

Barrier Functions



$$h \in C^1, \; S = \{x \mid h(x) \geq 0\}$$
$$\dot{h}(x) = \nabla h(x) \cdot \dot{x} \geq -\alpha h(x), \quad \forall x \in S$$

Approach safe set boundary with a *maximum rate*.

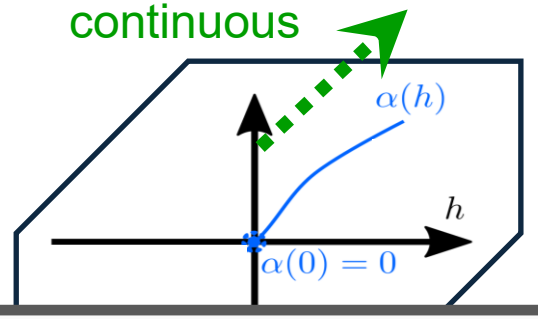
Safety with **CBFs**

Definition: *Control Barrier Functions*

The function h with regularity is a CBF if $\exists \alpha \in \mathcal{K}$ such that

$$\sup_{u \in \mathcal{U}} \dot{h}(x,u) = \nabla h(x)^T f(x,u) \geq -\alpha(h(x)), \quad \forall x \in S.$$

Monotonously increasing, continuous



Theorem: Let h be a CBF with a $\alpha \in \mathcal{K}$.

Any locally Lipschitz controller $u = k(x)$ satisfying

$$\nabla h(x)^T f(x, k(x)) \geq -\alpha(h(x)), \quad \forall x \in S$$

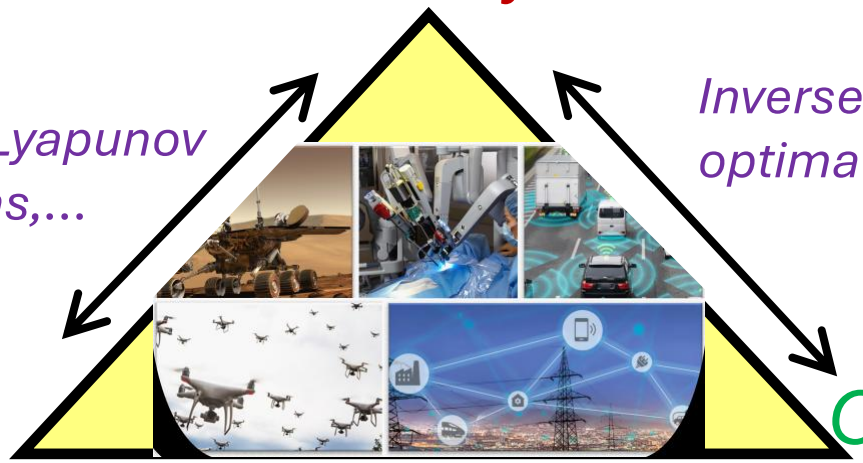
ensures that the system is safe.

A. D. Ames, et al., "*Control barrier functions: Theory and applications*." ECC, 2019.

Safety

Barrier-Lyapunov functions,...

Inverse optimality



Optimality

Stability

A modular solution for safe control

$$k(x) = \underset{u \in \mathcal{U}}{\operatorname{argmin}} \ \|u - k_{\text{nom}}(x)\|^2$$

$$\text{s.t.} \quad \nabla h(x)^T f(x,u) \geq -\alpha(h(x))$$

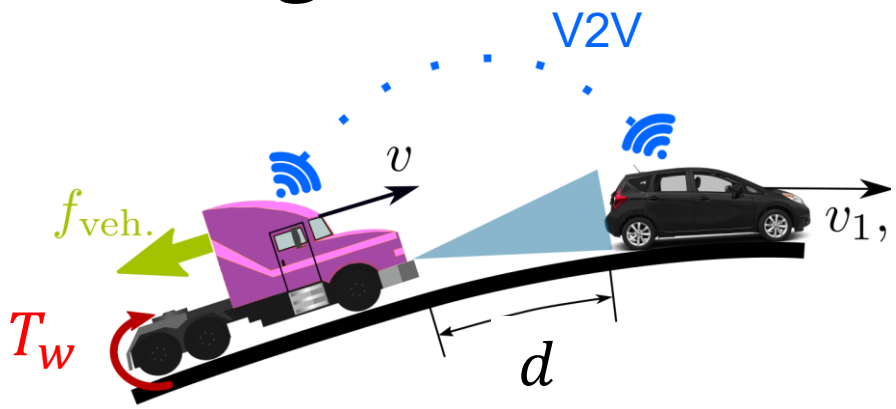
Nominal controller (Optimal control without constraint, RL, human operator…)

Safe ***car-following***

System: $x = [d, v]^T$

$$\dot{d} = v_1 - v$$

$$\dot{v} = \frac{T_w}{Rm_{\text{eff}}(gear)} - f_{\text{vehicle}}(v, gear)$$



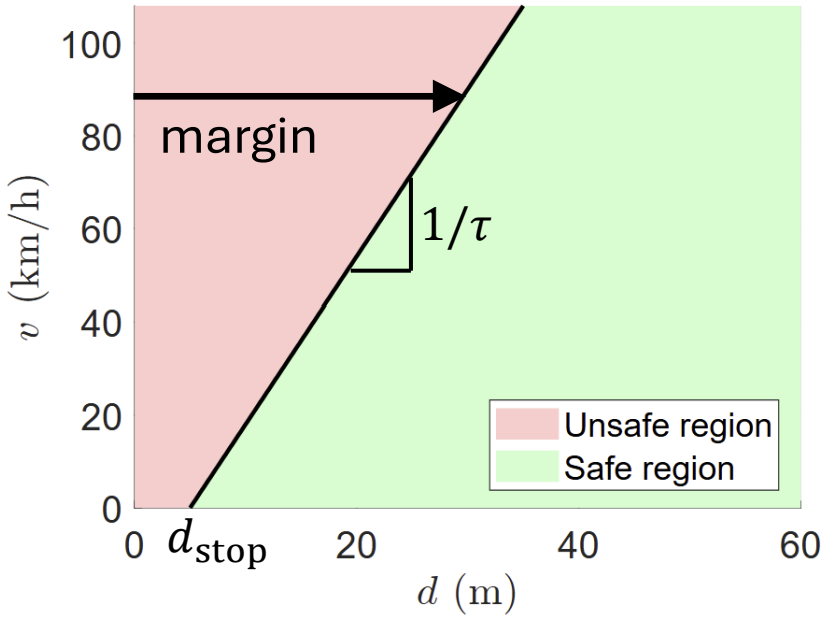
$$T_w \in [-T_{\text{maxBrake}}, T_{\text{maxDrive}}(gear)] = \mathbb{U}$$

Safety measure (desired safe set):

$$d \geq d_{\text{stop}} + \tau v$$

$$h(x) = d - d_{\text{stop}} + \tau v \geq 0$$

No CBF certification



Nominal controller: Human driver

```
%% ================= Environment parameters =================
para.g   = 9.81;  % m/s^2
para.rho = 1.204; % kg/m^3 air density at sea level, 20C
```

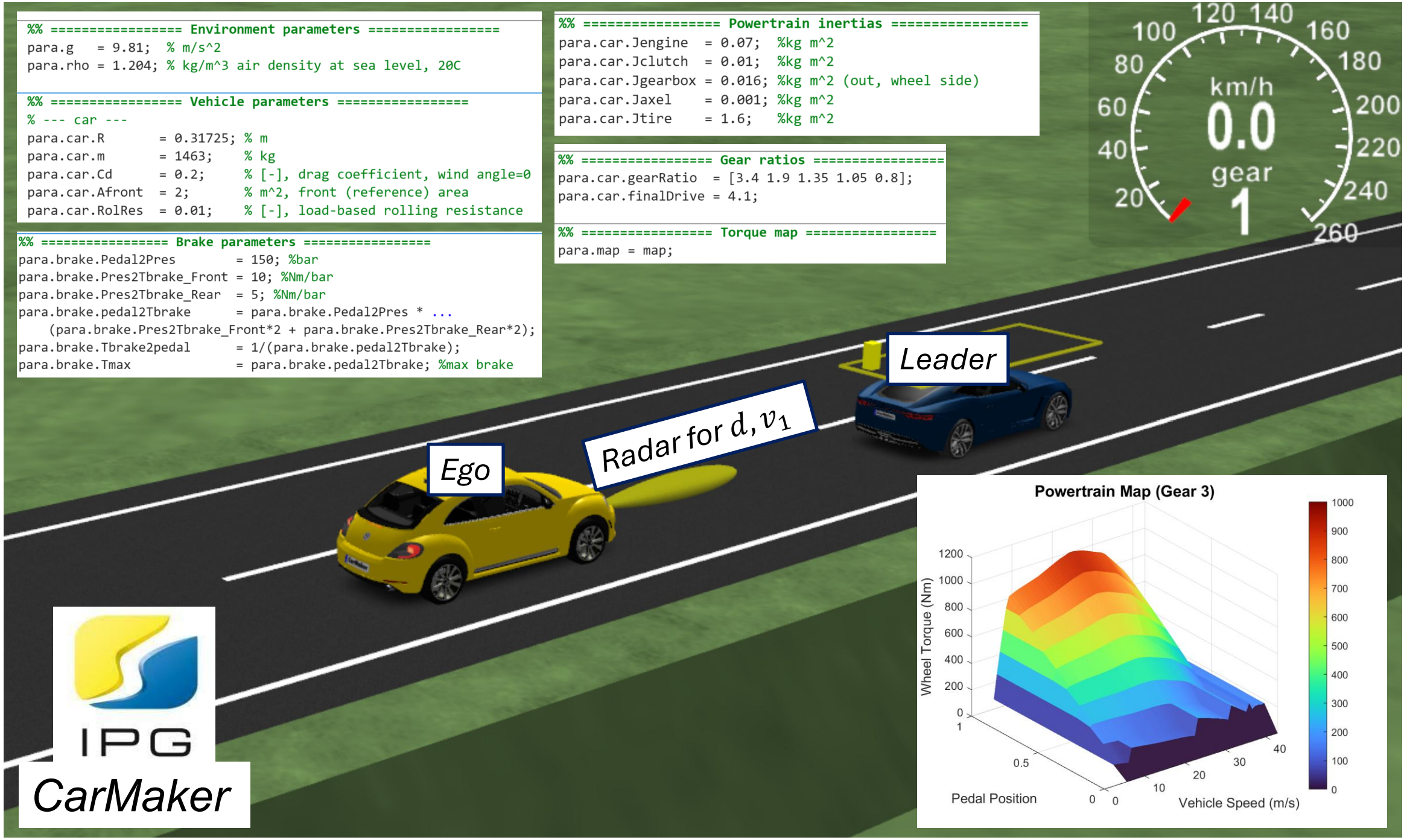
```
%% ================= Vehicle parameters =================
% --- car ---
para.car.R       = 0.31725; % m
para.car.m       = 1463;    % kg
para.car.Cd      = 0.2;     % [-], drag coefficient, wind angle=0
para.car.Afront  = 2;       % m^2, front (reference) area
para.car.RolRes  = 0.01;    % [-], load-based rolling resistance
```

```
%% ================= Brake parameters =================
para.brake.Pedal2Pres        = 150; %bar
para.brake.Pres2Tbrake_Front = 10; %Nm/bar
para.brake.Pres2Tbrake_Rear  = 5; %Nm/bar
para.brake.pedal2Tbrake      = para.brake.Pedal2Pres * ...
    (para.brake.Pres2Tbrake_Front*2 + para.brake.Pres2Tbrake_Rear*2);
para.brake.Tbrake2pedal      = 1/(para.brake.pedal2Tbrake);
para.brake.Tmax              = para.brake.pedal2Tbrake; %max brake
```

```
%% ================= Powertrain inertias =================
para.car.Jengine  = 0.07;  %kg m^2
para.car.Jclutch  = 0.01;  %kg m^2
para.car.Jgearbox = 0.016; %kg m^2 (out, wheel side)
para.car.Jaxel    = 0.001; %kg m^2
para.car.Jtire    = 1.6;   %kg m^2
```

```
%% ================= Gear ratios =================
para.car.gearRatio  = [3.4 1.9 1.35 1.05 0.8];
para.car.finalDrive = 4.1;
```

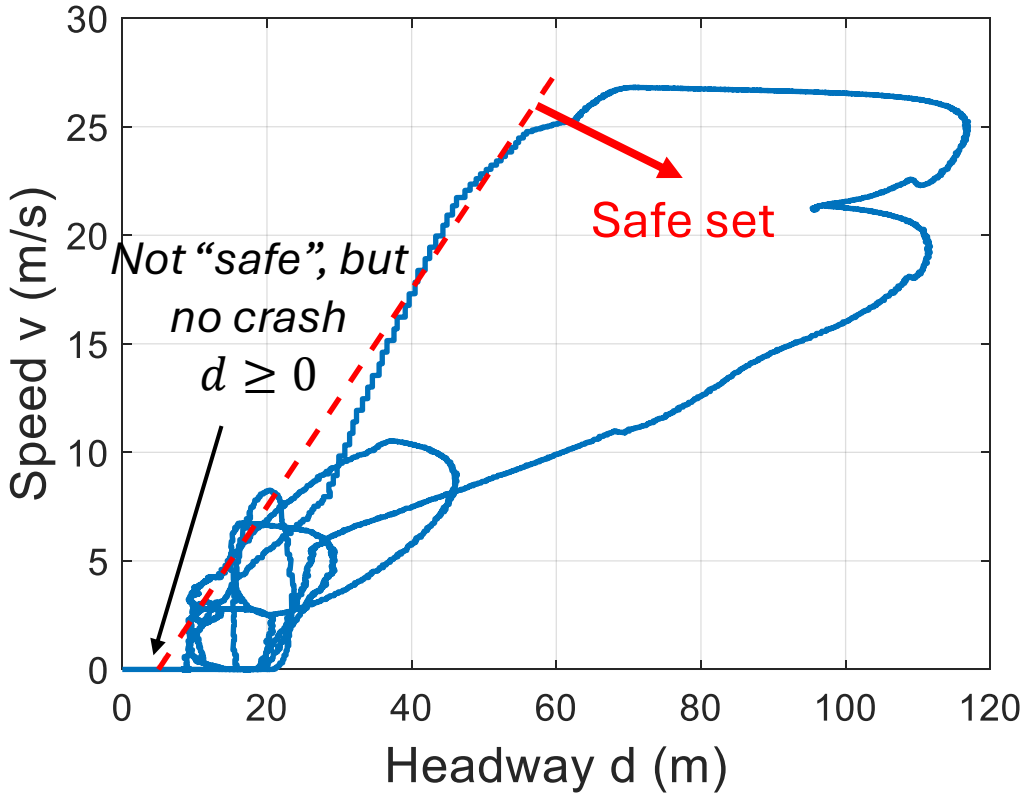
```
%% ================= Torque map =================
para.map = map;
```

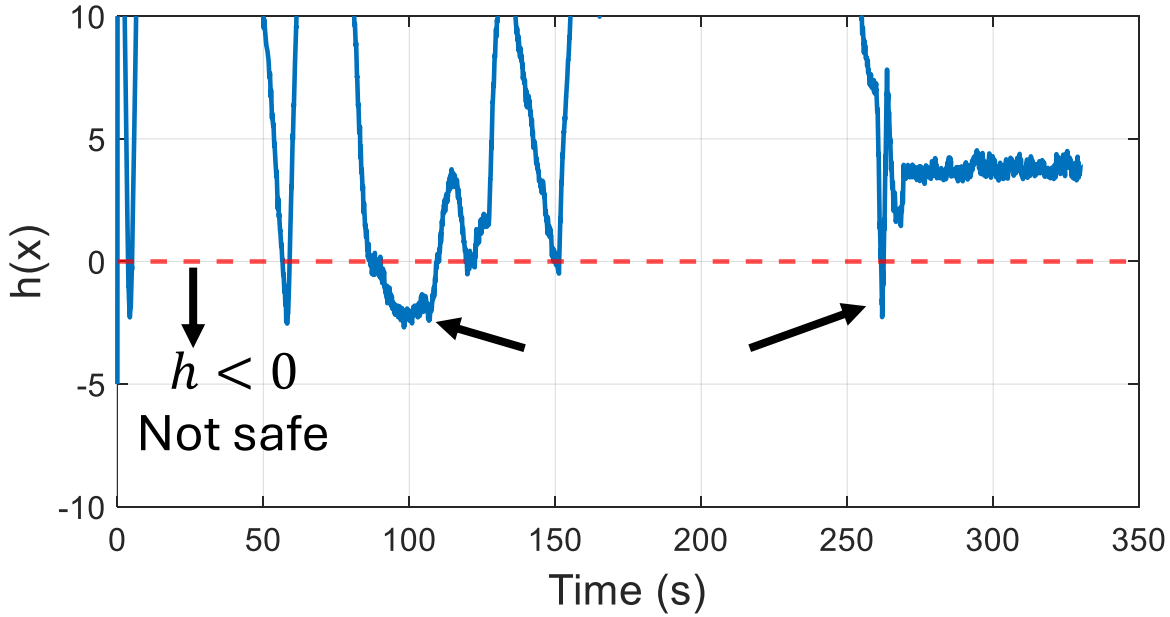



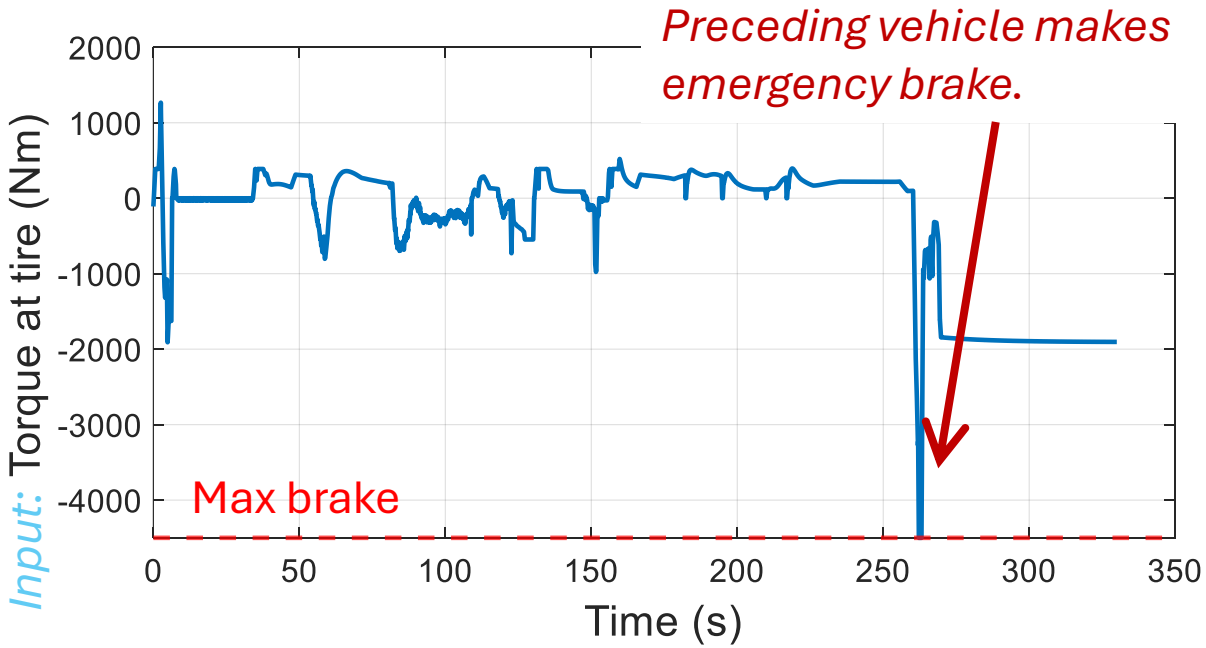
IPG CarMaker

Human driver *(CarMaker model)*



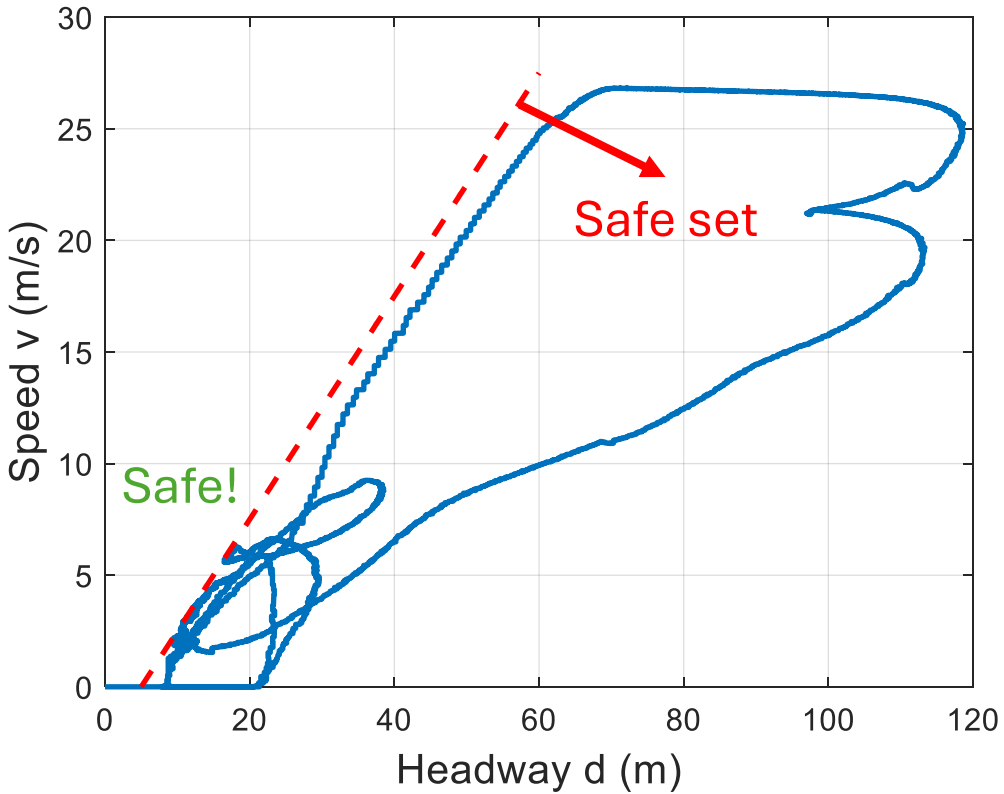


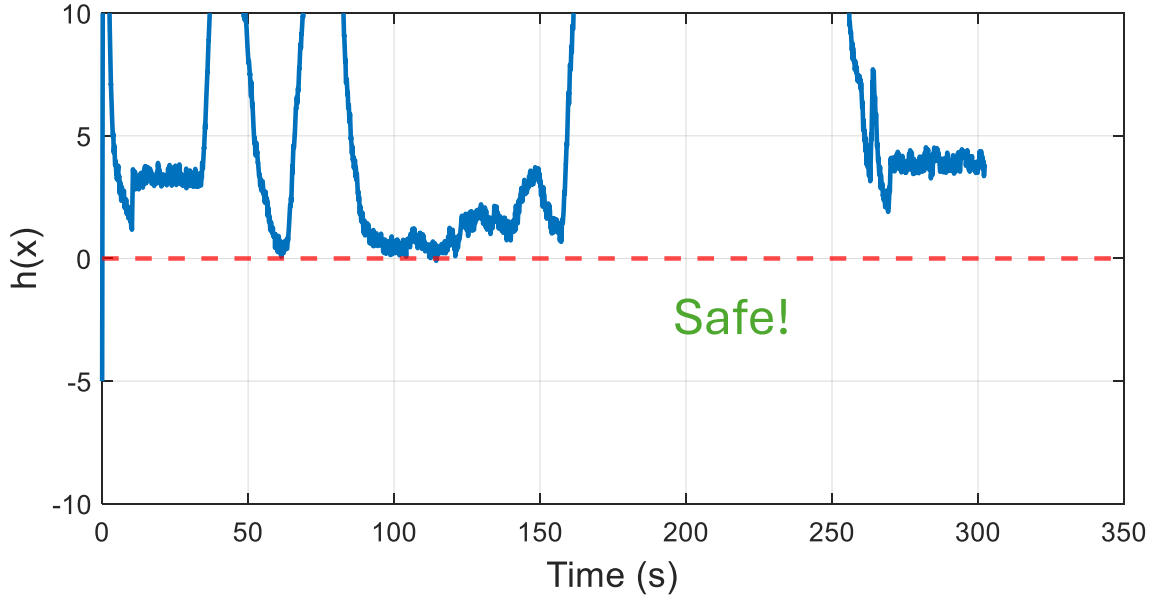


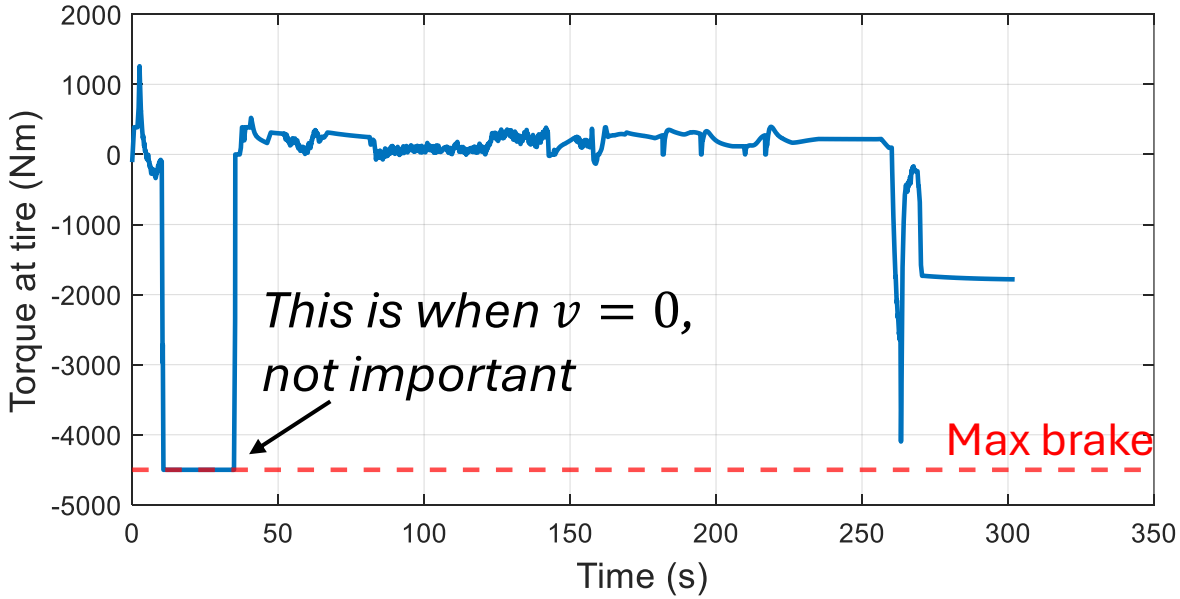


Safety filter *without CBF certification*









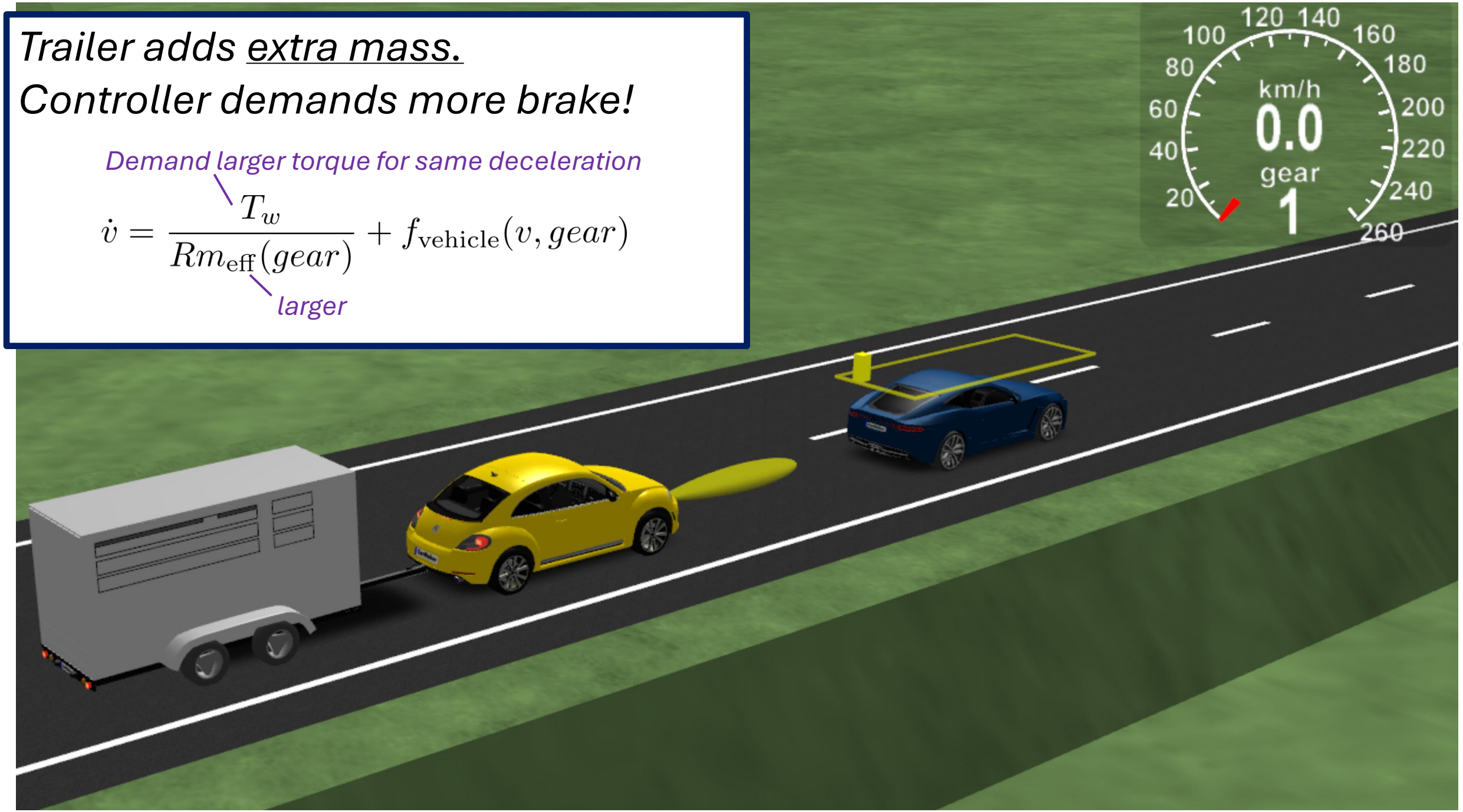
Trailer adds extra mass.

Controller demands more brake!

Demand larger torque for same deceleration

$$\dot{v} = \frac{T_w}{R m_{\text{eff}}(gear)} + f_{\text{vehicle}}(v, gear)$$

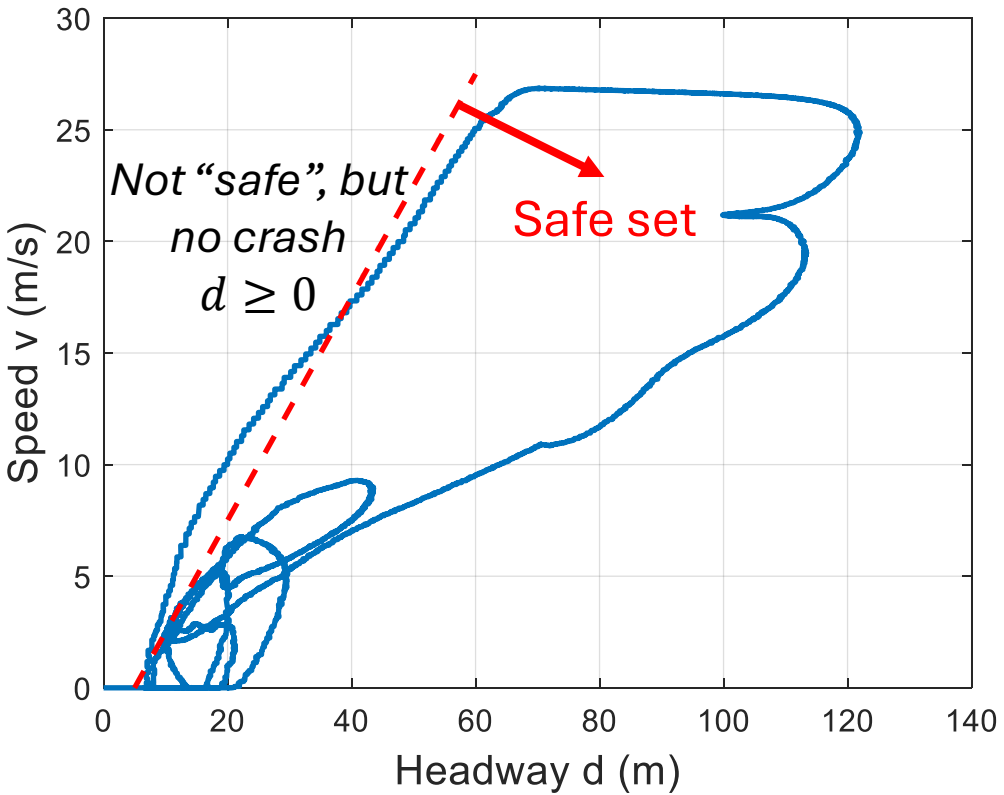
larger



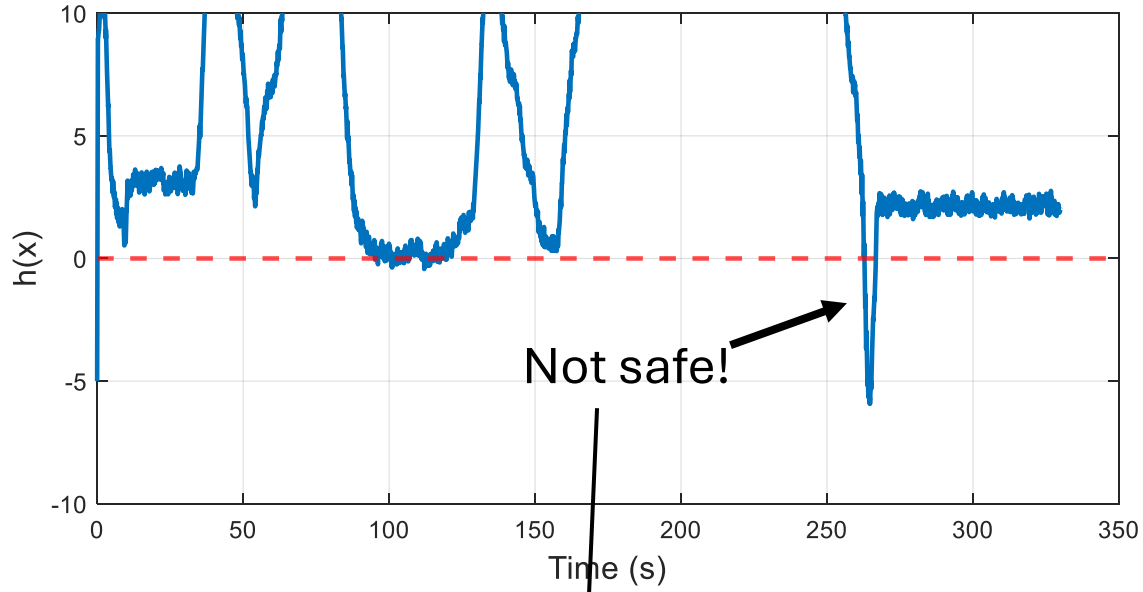
Safety filter *without CBF certification*

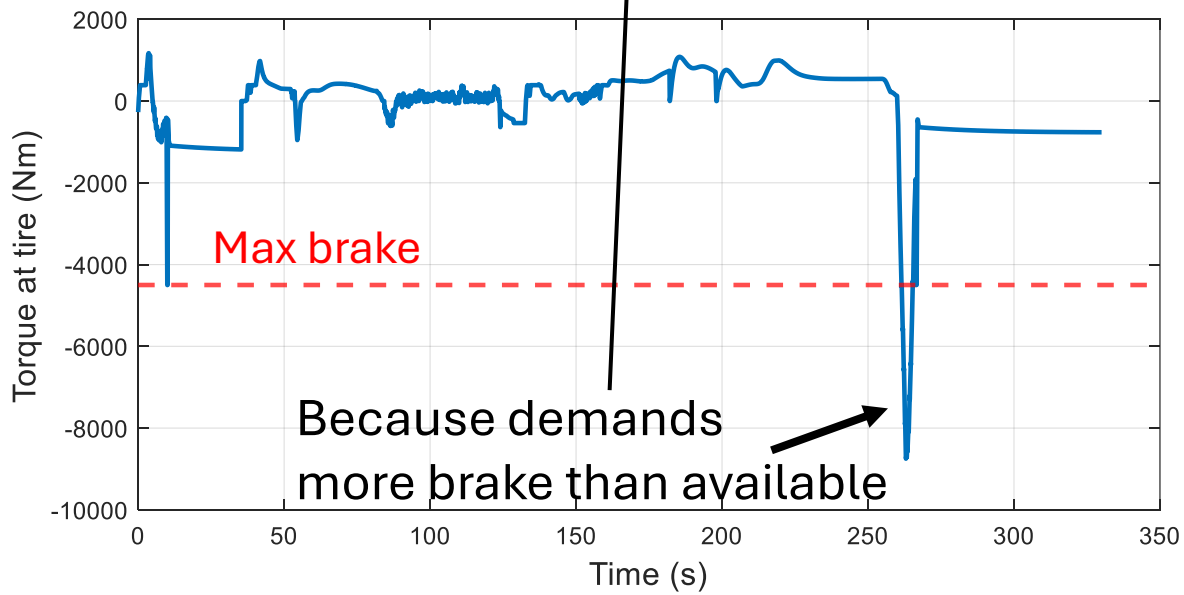






With trailer





Controlled Invariant Sets: *Backup construction*

Consider: $\mathcal{X}_0 = \{x \mid h(x) \geq 0\}$. *Without CBF certification:*

There may be a $x \in S$ such that $U_{\text{CBF}}(x) \triangleq \{u \in \mathbb{U} \mid \nabla h(x)^T f(x,u) \geq -\alpha(h(x))\} = \emptyset$.

Definition: *Controlled Invariance*

A set S is controlled invariant if

$\exists u(\cdot) \in \mathbb{U}$ such that $\phi_u(t; x_0) \in S, \forall t \geq 0 \; \forall x_0 \in S$.

- Backward set pruning.
- Solving Hamilton-Jacobi-Bellman equation.
- Semi-algebraic search (SOS).

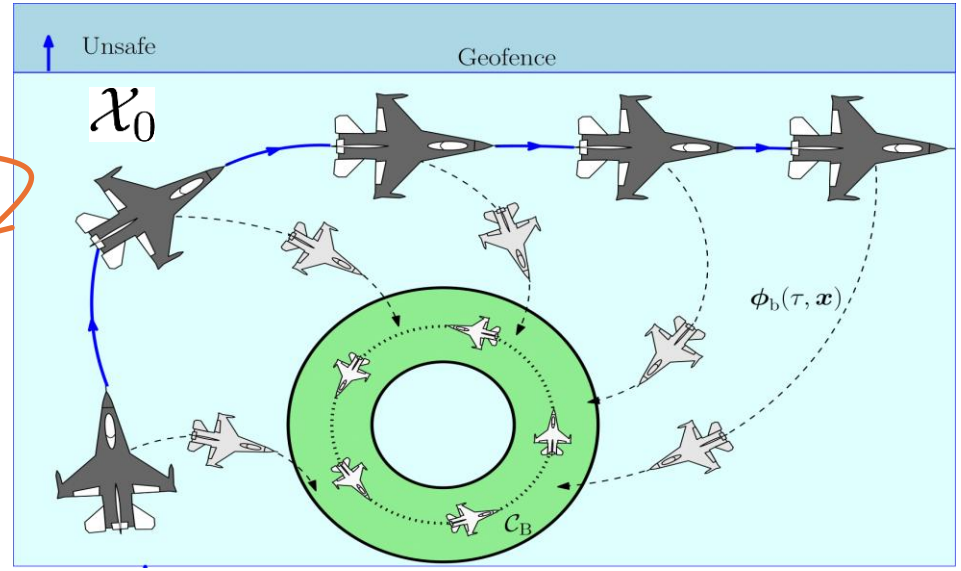
Constructing controlled invariant set using ***backup trajectories:***

Assume a Lipsch. cont. backup controller $u_b(\cdot)$ exists: $\phi_b(t; x_0)$

$$S_I^T \triangleq \left\{ x \;\middle|\; \begin{array}{l} h(\phi_b(\tau; x_0)) \geq 0, \; \forall \tau \in [0, T] \\ h_b(\phi_b(T; x_0)) \geq 0 \end{array} \right\}$$

is controlled invariant.if terminal set $S_b = \{x \in \mathcal{X}_0 \mid h_b(x) \geq 0\}$

is controlled invariant too.[*]



[*]D. E. J. van Wijk et al., "Safety-Critical Control with Bounded Inputs: A Closed-Form Solution for Backup Control Barrier Functions", 2026

Controlled Invariant Sets – CBF relationship

Controlled invariant sets do not necessarily provide a CBF!

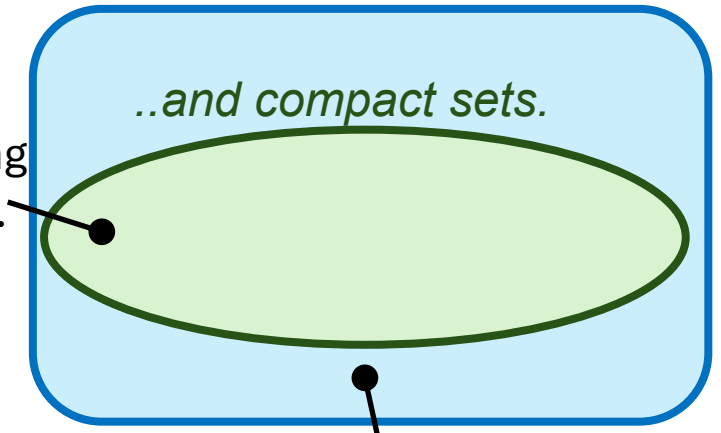
$$h \text{ is a CBF} \implies S = \{x \mid h(x) \geq 0\} \text{ is a controlled invariant set.}$$

$\Longleftarrow$? ✓

Converse theorem: Yes, if S is compact.

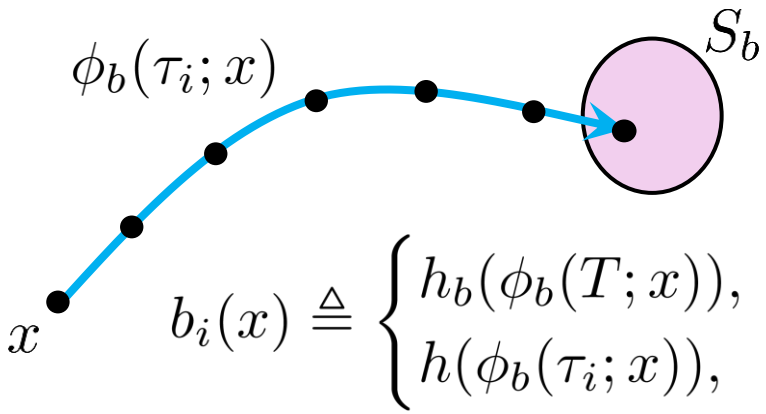
P. Mestres, J. Cortez, "*Converse Theorems for Certificates of Safety and Stability*" IEEE TAC 2025.

Controlled invariant sets (with regular h)



Computing S_I^T needs $\tau \in [0, T]$ discretizing in N.

$$\widehat{S} = \{x \mid \min b_i(x) \geq 0\}$$



$$b_i(x) \triangleq \begin{cases} h_b(\phi_b(T;x)), & i = N \\ h(\phi_b(\tau_i;x)), & \text{else} \end{cases}$$

Theorem: *Converse safety for backup*

Let $\tilde{S} = \{x \mid \text{softmin}_\theta \{\min b_i(x)\} \geq 0\}$.

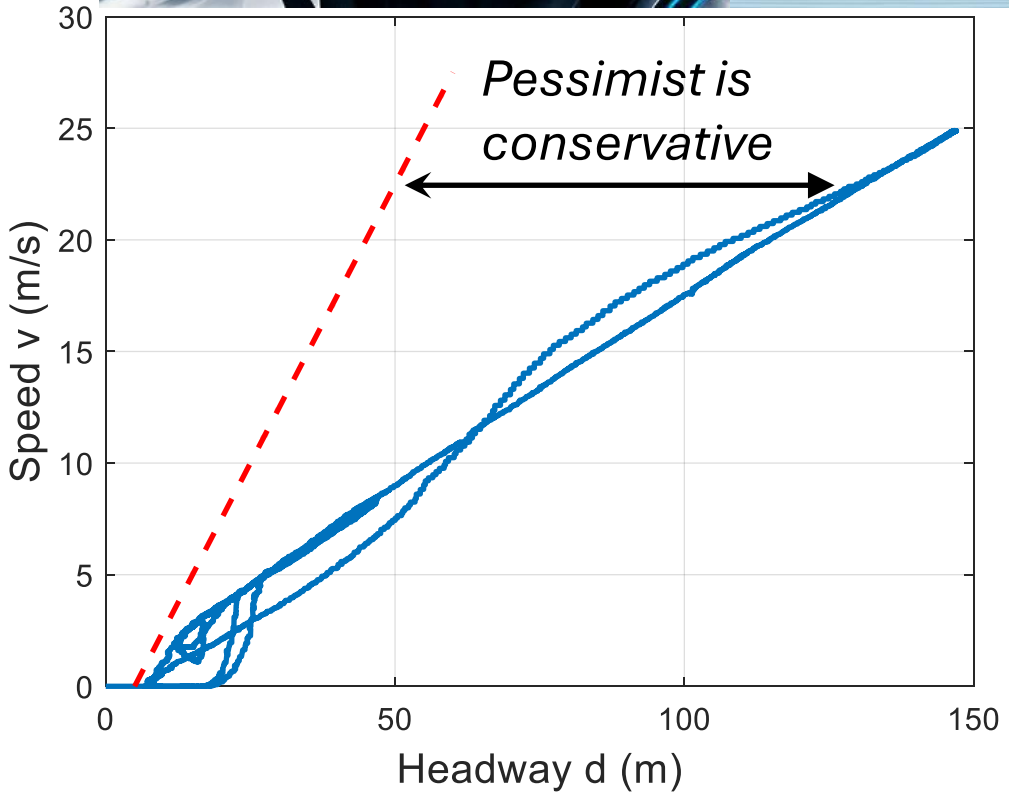
If S_b is compact and *strictly* controlled invariant, then softmin function is a CBF.

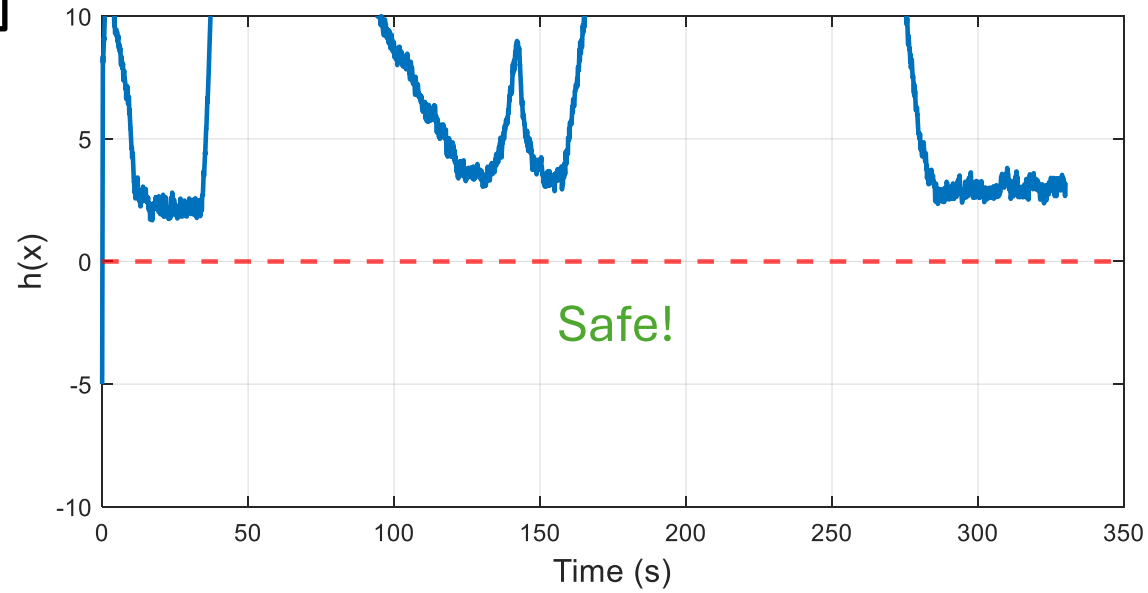
A. Alan, B. De Schutter, *"Uniform Feasibility for Smoothed Backup CBF"*, ECC 2026

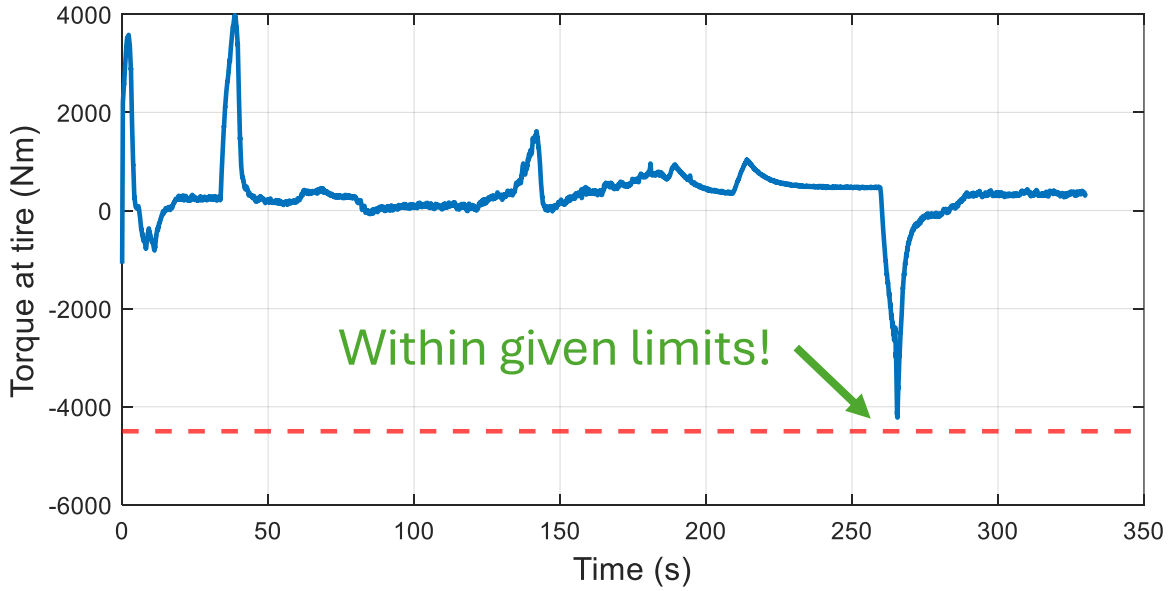
Safety filter *with backup CBF verification*



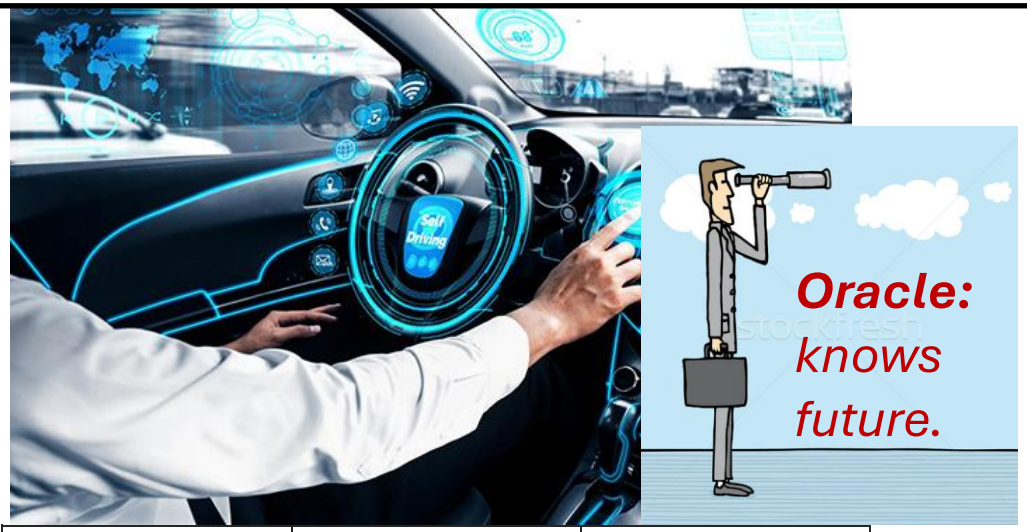


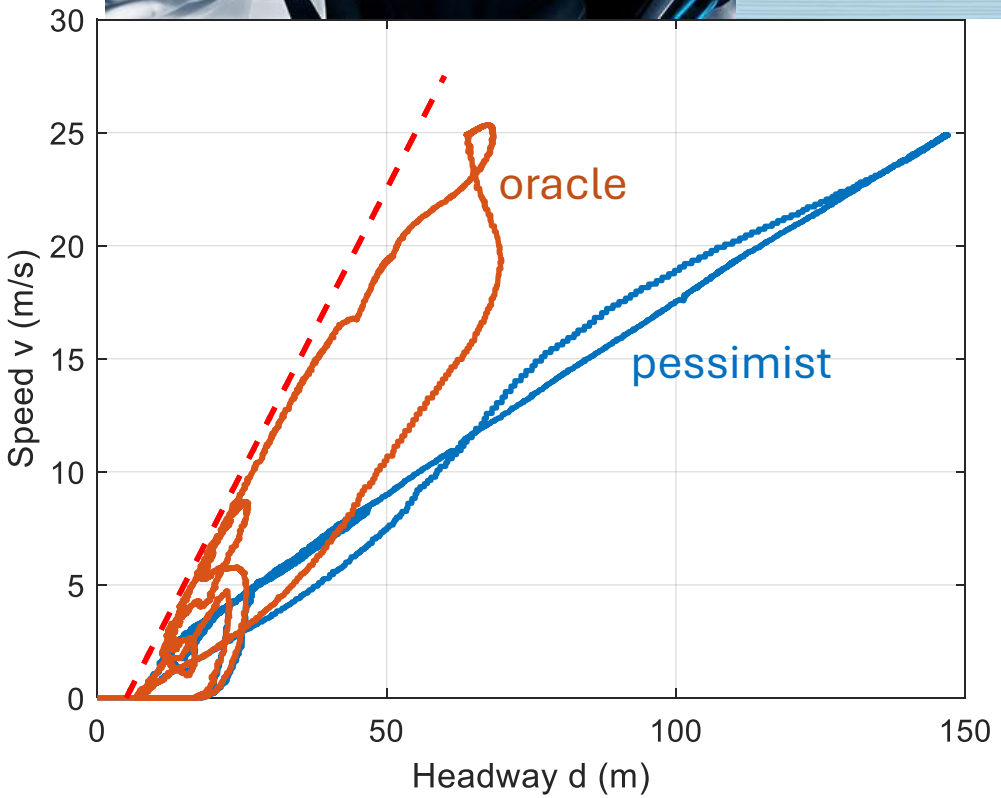


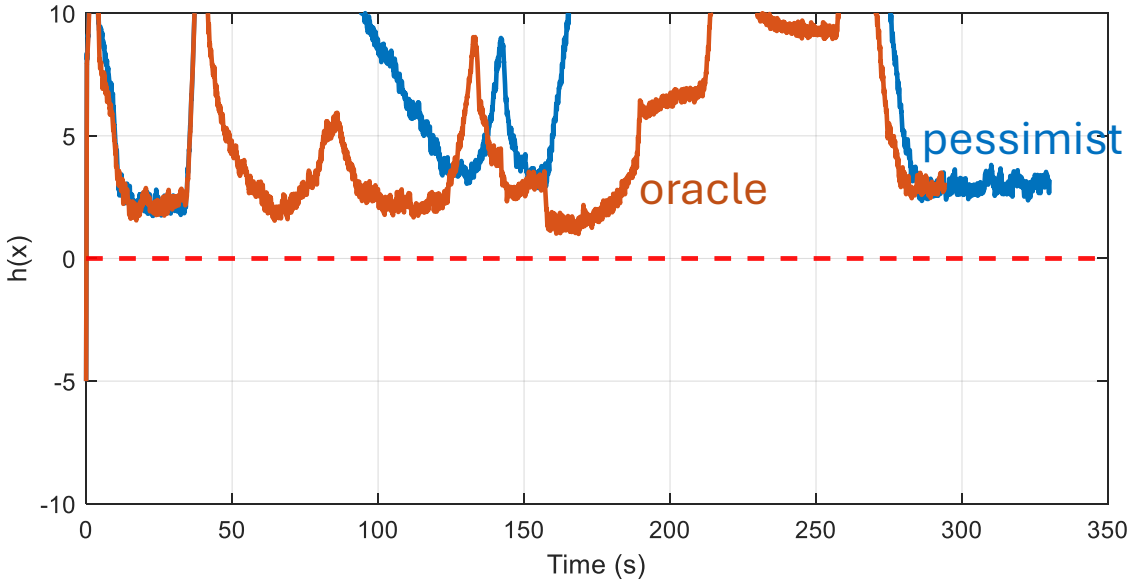


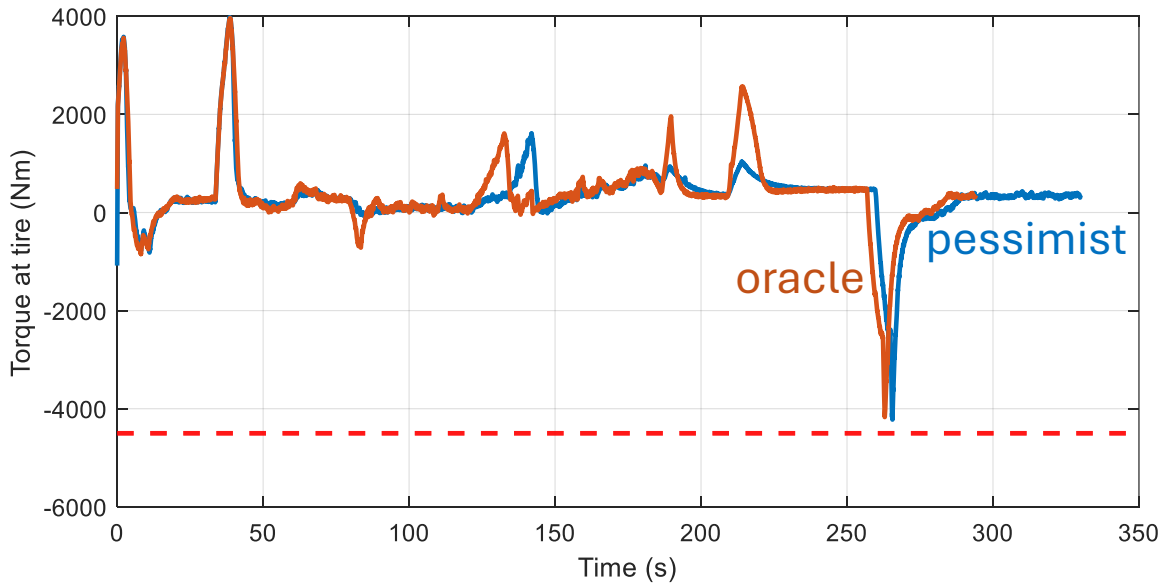


Safety filter *with backup CBF verification*

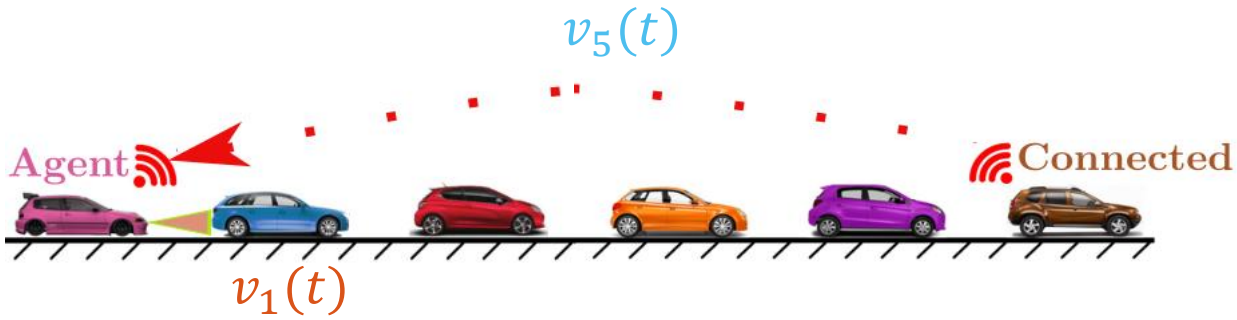


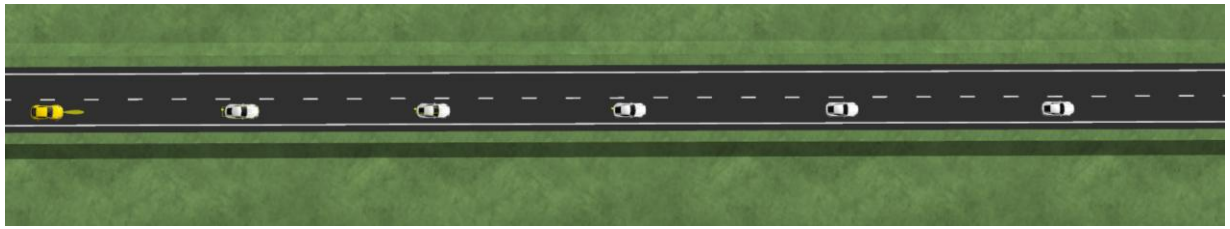






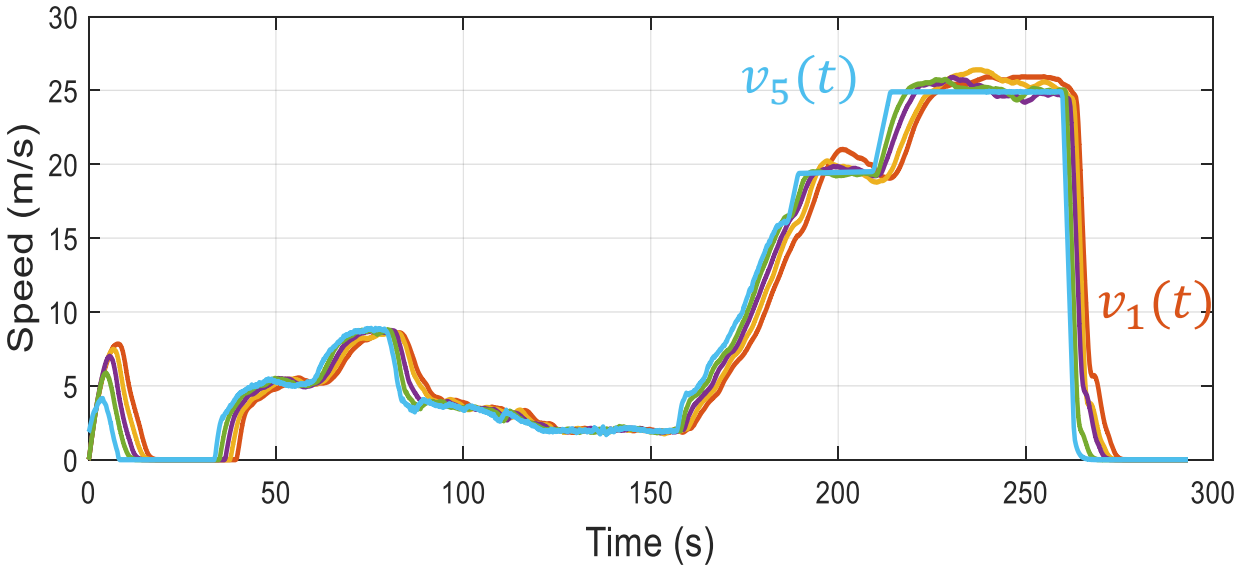
Can we estimate the preceding vehicles speed using traffic estimator?

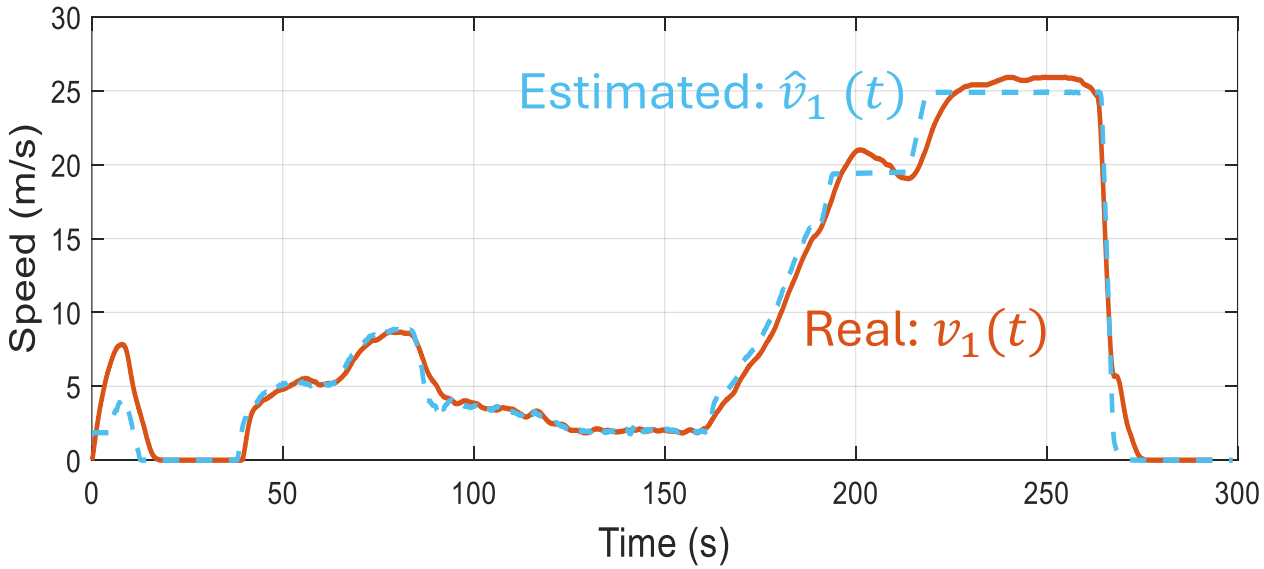




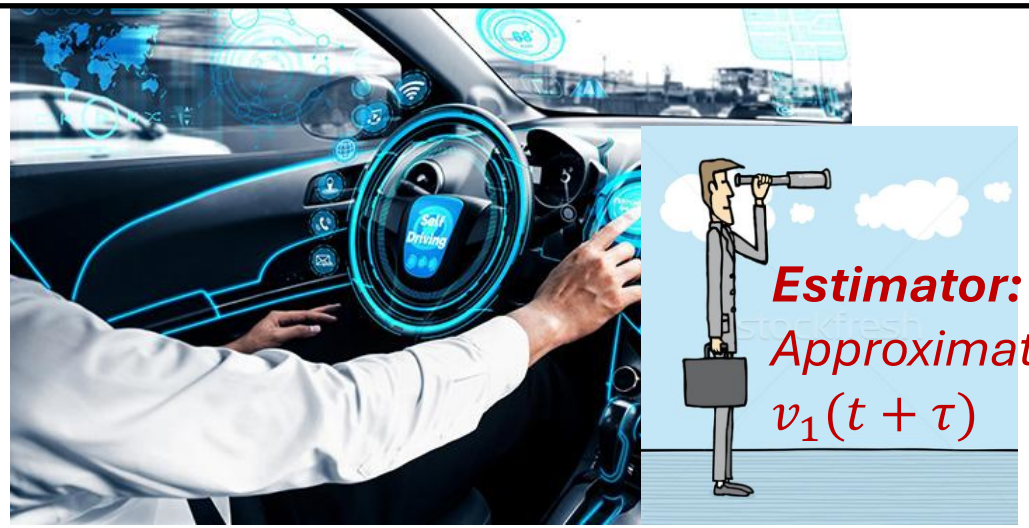
Estimator: $\hat{v}_1(t) = v_5(t-\theta)$

Shift the broadcasted speed backwards θ seconds

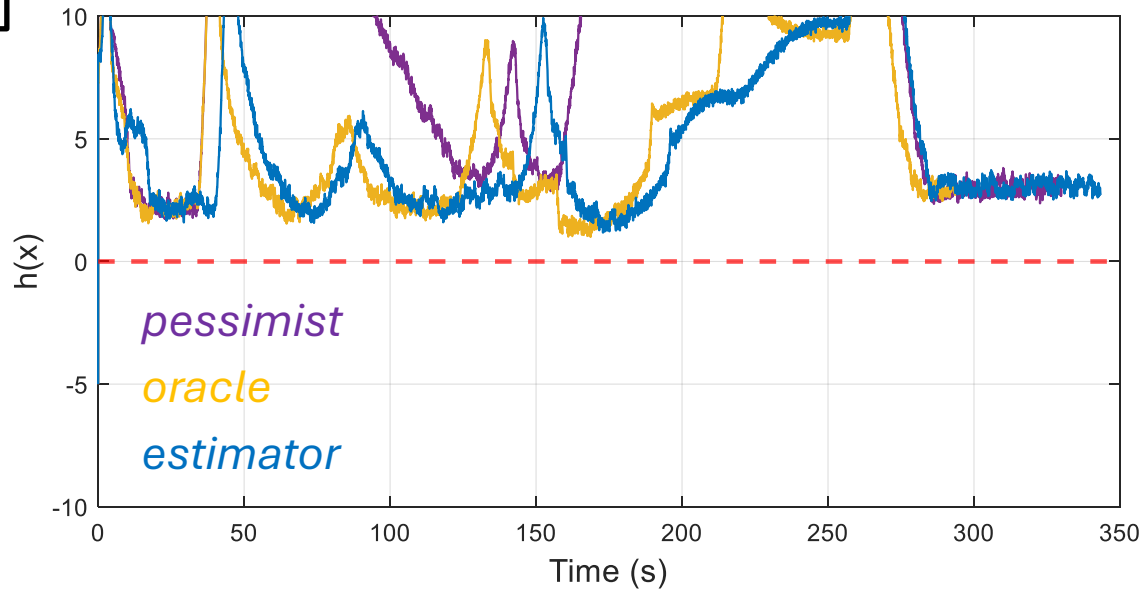


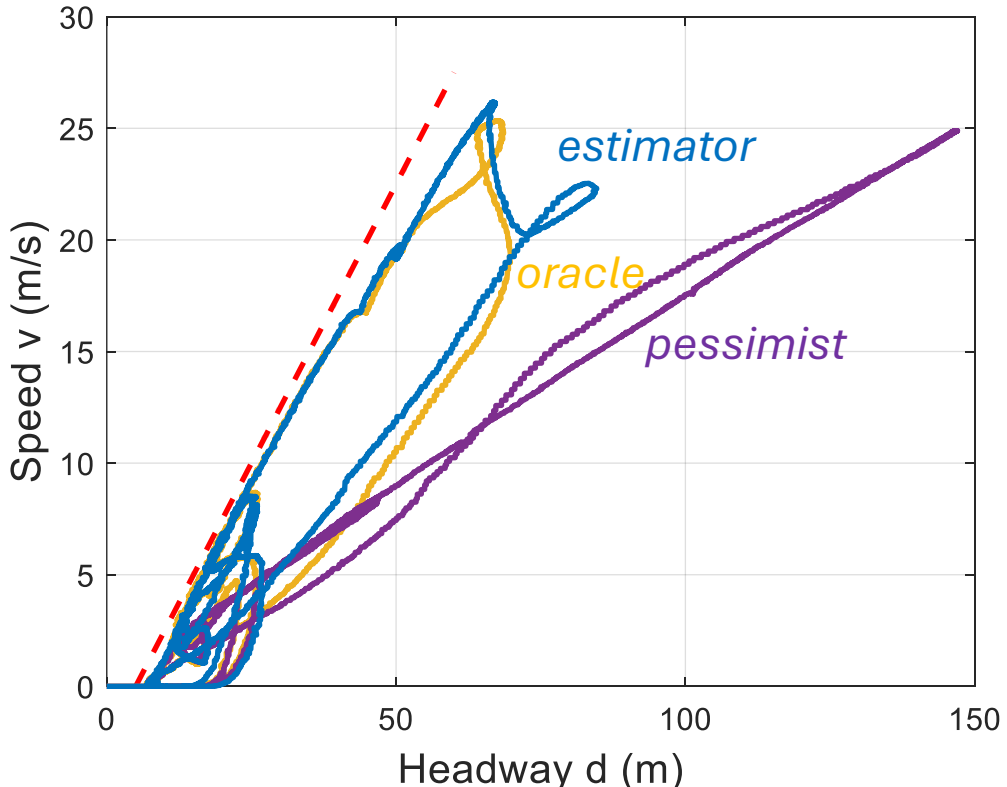


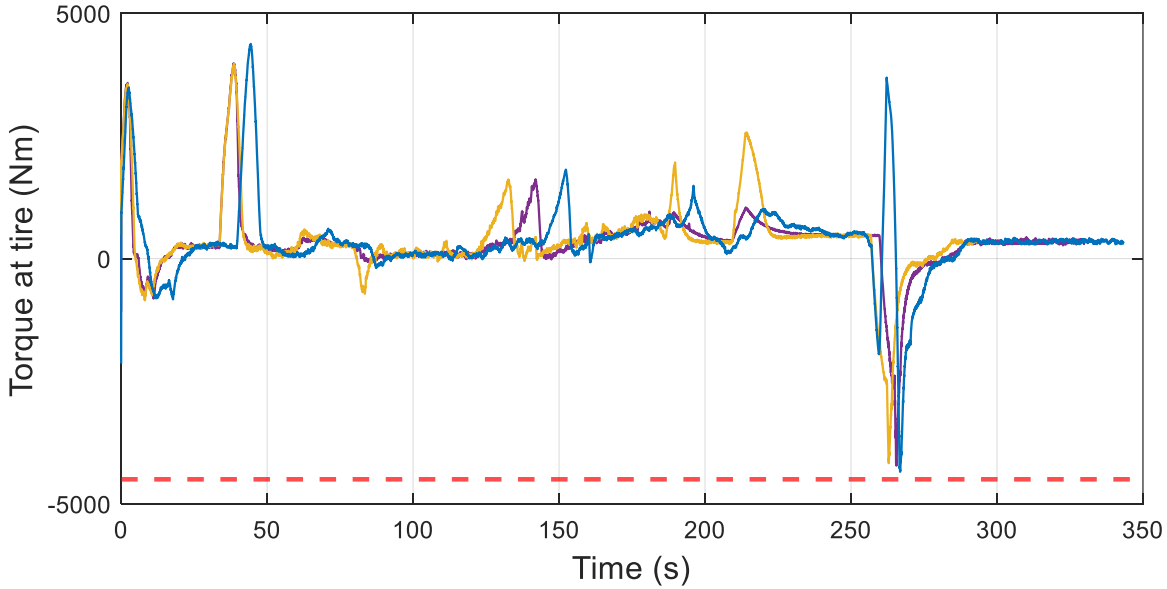
Safety filter *with backup CBF verification*



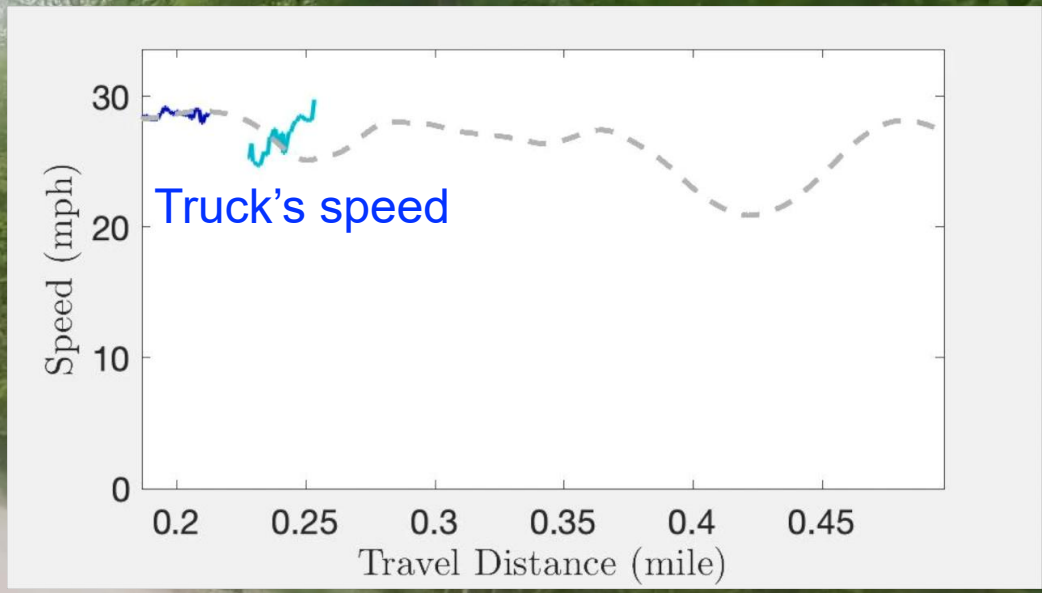
Estimator:
Approximates
$v_1(t+\tau)$







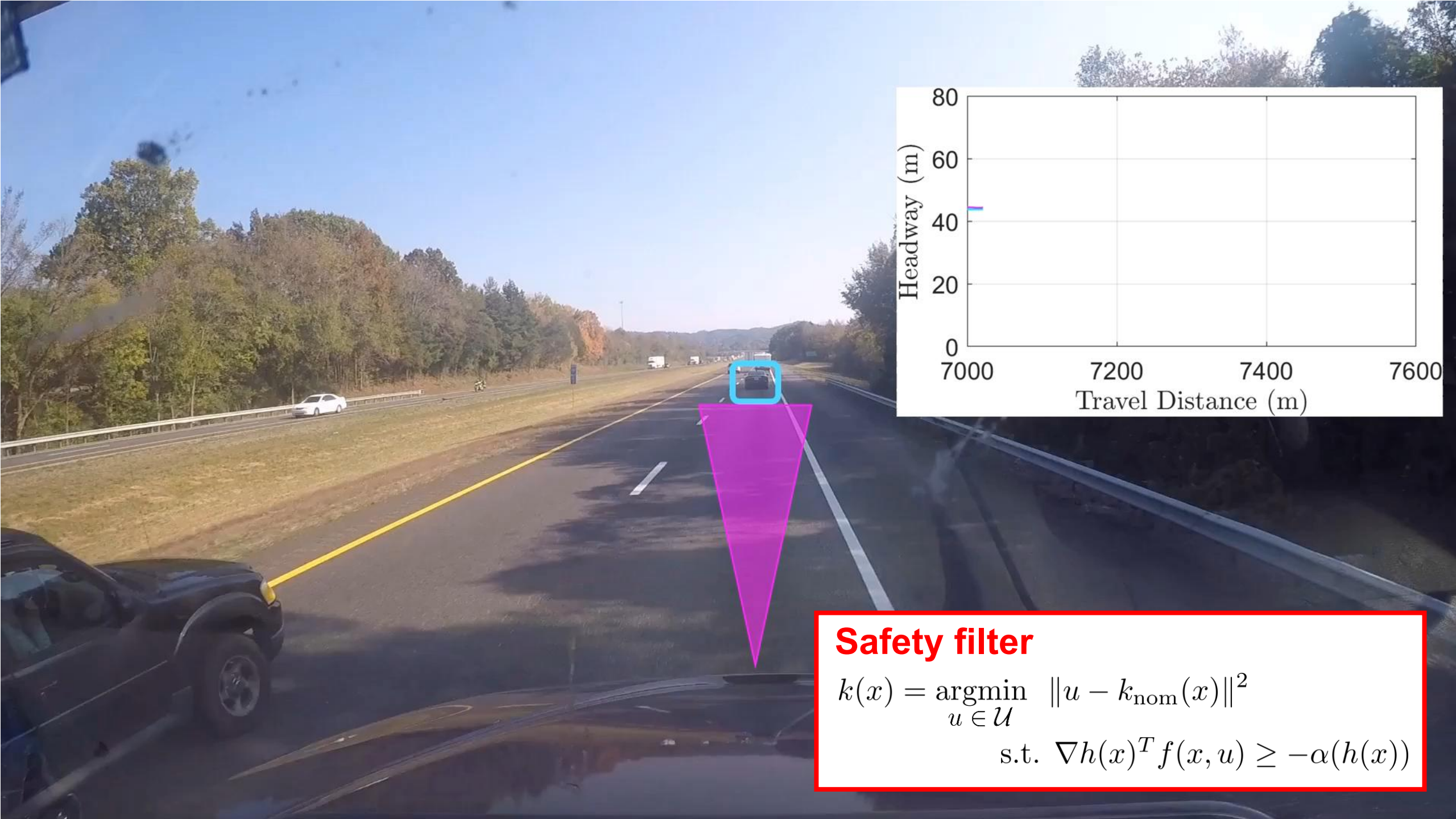

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Safety filter

$$k(x) = \underset{u \in \mathcal{U}}{\operatorname{argmin}} \ \|u - k_{\text{nom}}(x)\|^2$$

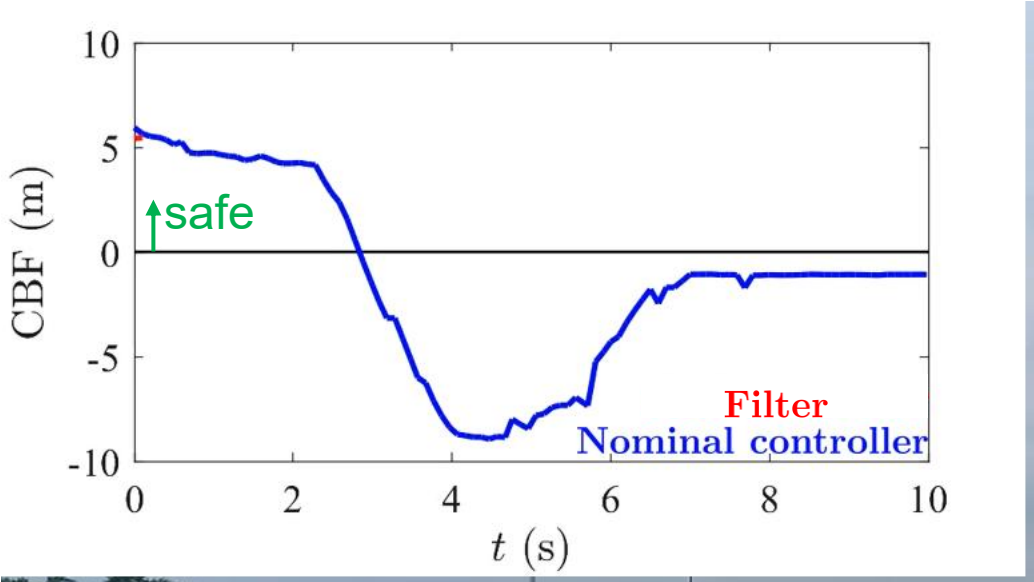
$$\text{s.t. } \nabla h(x)^T f(x,u) \geq -\alpha(h(x))$$



Safety filter

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$$\text{s.t.} \ \nabla h(x)^T f(x,u) \geq -\alpha(h(x))$$





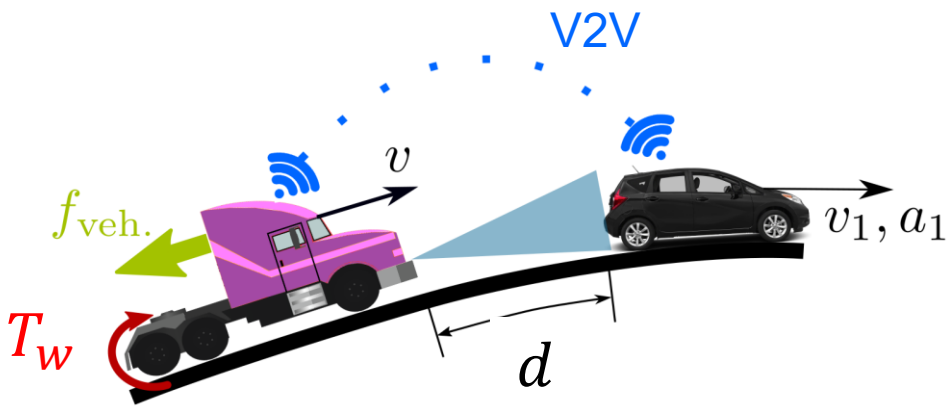
Safety filter

$$k(x) = \underset{u \in \mathcal{U}}{\operatorname{argmin}} \ \|u - k_{\text{nom}}(x)\|^2$$

$$\text{s.t.} \ \ \nabla h(x)^T f(x,u) \geq -\alpha(h(x))$$

Uncertainties are inherent in control design

Theoretical Model

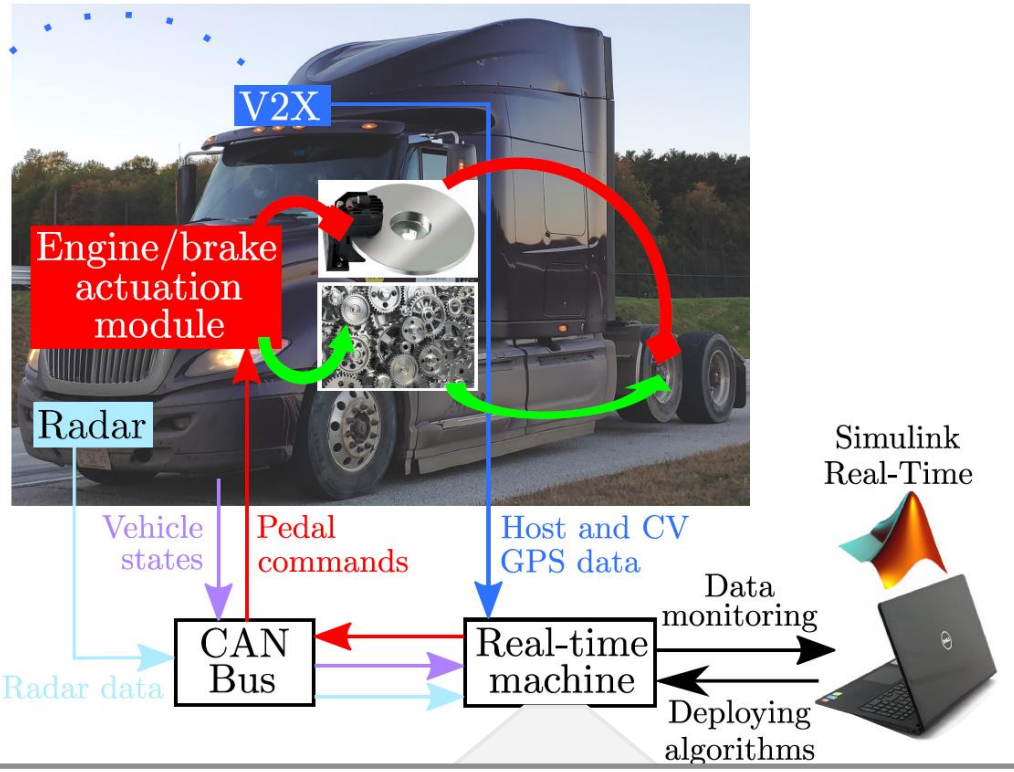


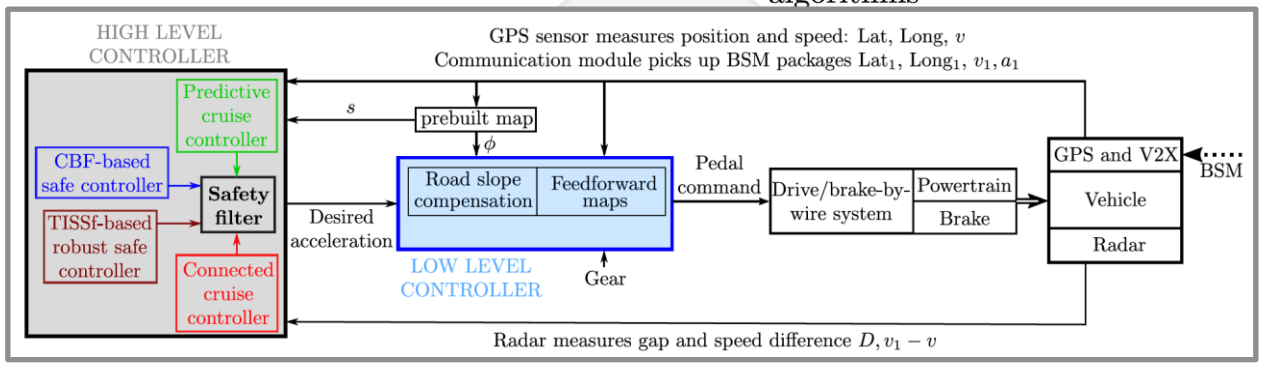
$$\dot{d} = v_1 - v$$

$$\dot{v} = \frac{T_w}{Rm_{\text{eff}}(gear)} + f_{\text{vehicle}}(v, gear)$$

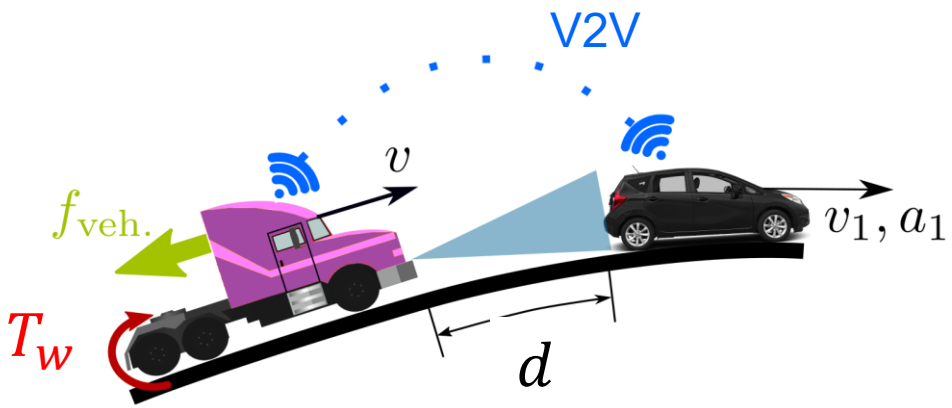
Ideal powertrain/brake

Real Implementation



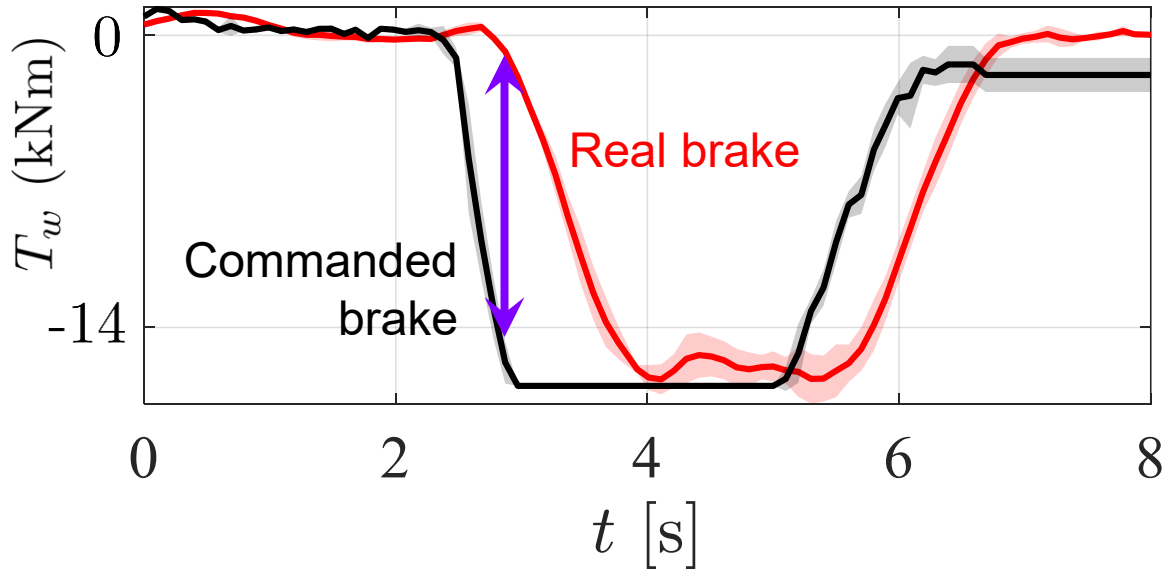


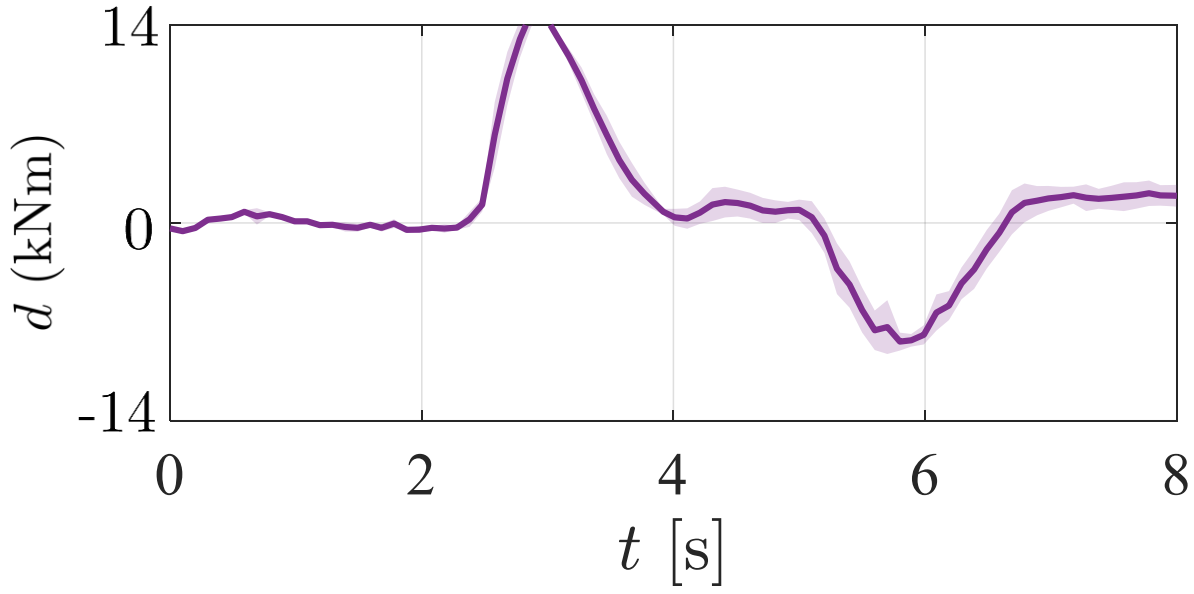
Uncertainties are inherent in control design



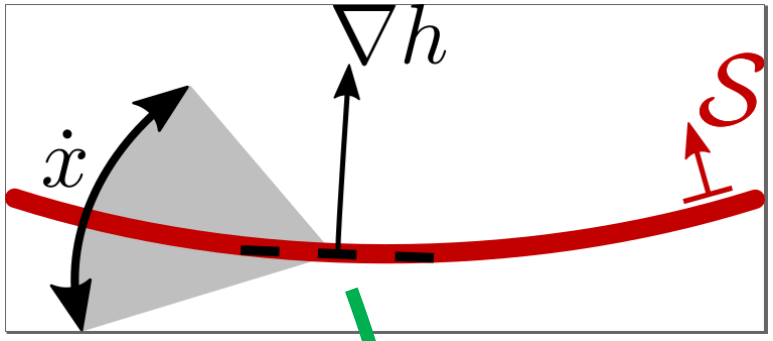
$$\dot{d} = v_1 - v$$

$$\dot{v} = \frac{T_w + d(t)}{Rm_{\text{eff}}(gear)} + f_{\text{vehicle}}(v, gear)$$





Robust methods *in CBF*



Deterministic bounds $d \in \mathcal{D}$

Distributed $d \sim \mathcal{N}(\mu, \sigma^2)$

Chance-constraint CBF

C. Wang, et al., "*Chance constraint robust control with control barrier functions*." ACC, 2021.

$$\max_{u \in \mathbb{U}} \min_{d \sim \mathcal{N}(\mu, \sigma^2)} \dot{h}(x, u, d) \geq -\alpha(h(x))$$

Robust CBF

M. Jankovic, "*Robust control barrier functions for constrained stabilization of nonlinear systems.*" Automatica, 2018.

Cancel worst-case: $\max_{u \in \mathbb{U}} \min_{d \in \mathcal{D}} \dot{h}(x, u, d) \geq -\alpha(h(x))$

Input-to-state safety

Graceful relaxation: $\max_{u \in \mathbb{U}} \min_{d \in \mathcal{D}} \dot{h}(x, u, d) \geq -\alpha(h(x)) - \mu(\|d\|)$

Input-to-State Safety

Disturbed control system $\dot{x} = f(x) + g(x)(u + d(t)),$

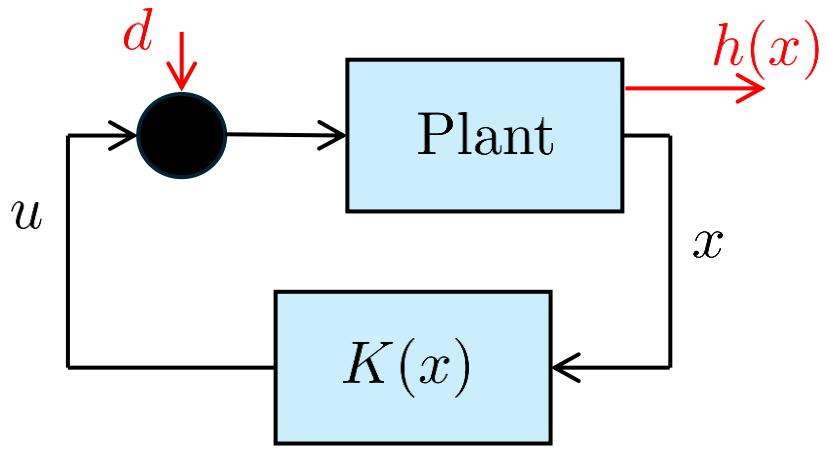
Bounded disturbance $d(\cdot) \in \mathcal{D}_{\bar{d}} = \left\{ d \in L_\infty \mid \|d\|_\infty \leq \bar{d} \right\}.$

Definition: *Input-to-State Safety*

Let $S = \{x \mid h(x) \geq 0\}$ with $h : \mathbb{R}^n \to \mathbb{R},\ S \subseteq \mathcal{X}_0.$

System is input-to-state safe if $\exists \sigma \in \mathcal{K}$ such that

$$\forall x_0 \in S,\ h(\phi(t; x_0, d)) \geq -\sigma(\|d\|),\ \ \forall d(\cdot) \in \mathcal{D}_{\bar{d}},\ \forall t \geq 0.$$



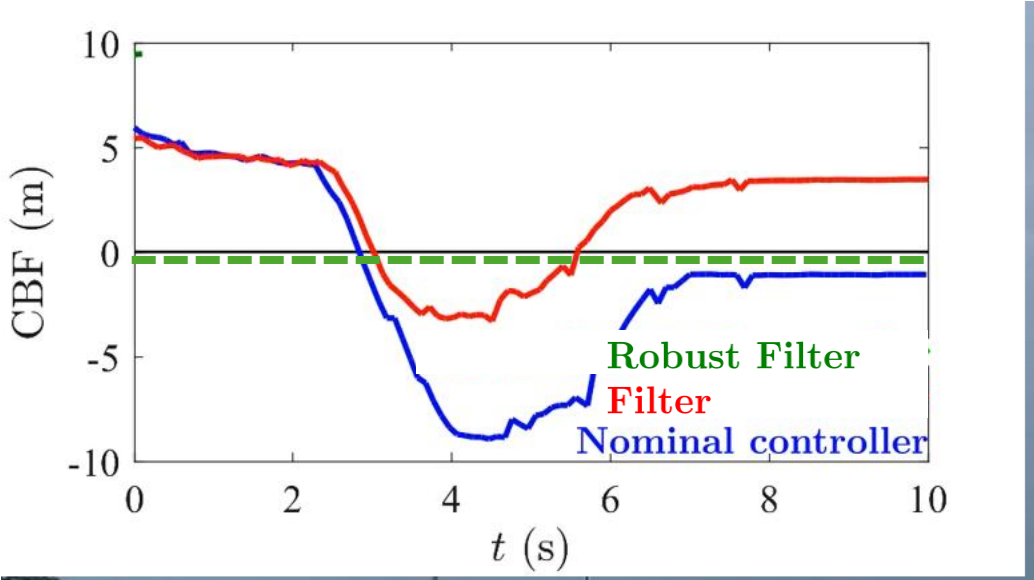
Theorem: Sufficiency

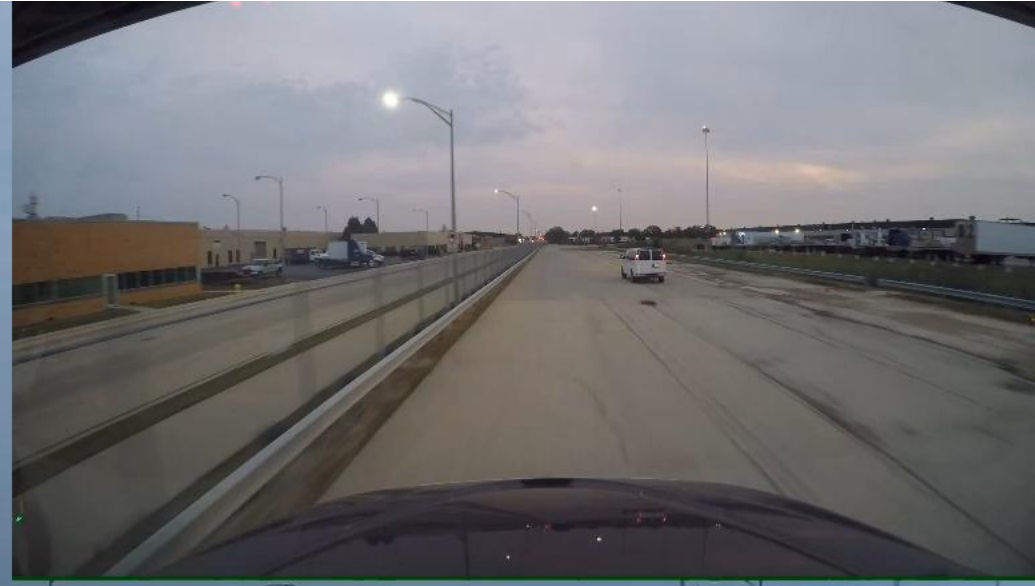
If $\exists \alpha \in \mathcal{K},\ \epsilon > 0$ such that $\forall x \in \mathbb{R}^n$:

$$\dot{h}(x) \geq -\alpha(h(x)) + \|\nabla h(x)^T g(x)\|^2 \epsilon,$$

then system is ISSf with $\sigma(\|d\|) = -\alpha^{-1}\left(-\frac{\|d\|^2}{4\epsilon}\right) \in \mathcal{K}.$

A. Alan, et al., "CBF *and ISSf with application to automated vehicles*", IEEE TCST, 31-6, 2023.





Robust (ISSf) safety filter

$$k(x) = \underset{u \in \mathcal{U}}{\operatorname{argmin}} \ \|u - k_{\text{nom}}(x)\|^2$$

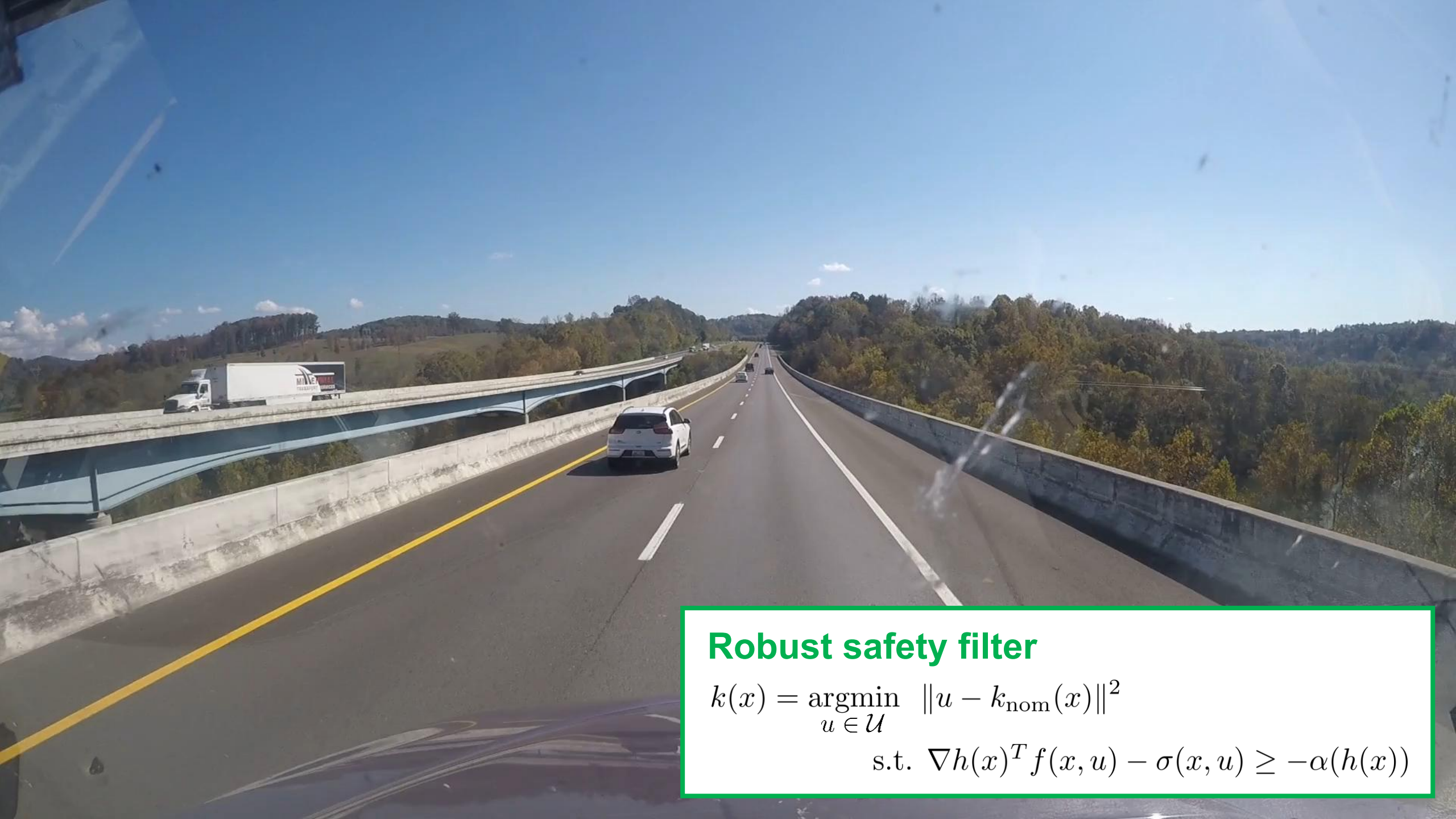
$$\text{s.t. } \nabla h(x)^T f(x,u) \geq -\alpha(h(x)) + \|\nabla h(x)^T g(x)\|^2 \epsilon$$

There are many ways to "robustify" CBF

Sufficient condition: $L_f h(x,u) - \sigma(x,u) \geq -\alpha(h(x))$

		Method Summary	$\sigma(t,x,u)$
RCBF	[7]	Bounded uncertainty: $\|\tilde{f}(t,x)\| \leq p$.	$\|\nabla h(x)\| p$
	[28]	Bounded uncertainty: $\|\tilde{f}(t,x)\| \leq p$, continuous non-increasing κ with $\kappa(0)=1$	$\kappa(h(x))\|\nabla h(x)\| p$
	[15]	$\tilde{f}$ is a convex hull of functions $\psi_i(x), i=1,\cdots,q$, $\tilde{g}$ is a convex hull of functions $\rho_i(x), i=1,\cdots,q$,	$-\min_{i\in\{1,..,q\}} \nabla h(x)\phi_i(x)$ $-\min_{i\in\{1,..,q\}} \nabla h(x)\rho_i(x)u$
	[12]	$[\tilde{f}, (\tilde{g}\ \mathrm{diag}(u))^\top]^\top = \psi(x,u)\theta$, and $\exists A, b$ s.t. $A\theta \leq b$	$\inf_{A\theta\leq b} \nabla h(x)\psi(x,u)\theta$
	[13]	Sector bounded nonlinear perturbation at input, i.e., $\exists \alpha, \beta$ defining $g_s = (\alpha+\beta)g/2$ and $\theta = (\beta-\alpha)/(\beta+\alpha)$.	$(L_g h(x) - L_{g_s} h(x))\, u + \theta\|u\|\|L_{g_s} h(x)\|$
	[8]	$\hat{f}$ estimates $\tilde{f}$ and b_d defines an error band.	$-\nabla h(x)\hat{f}(x) + b_d(t)$
	[9]	$b(t,x) = L_{\tilde{f}} h(t,x)$ is Lipschitz in t (constant L_b), $\hat{b}$ estimates b and k_b is estimation gain.	$-\hat{b}(t,x) + L_b/k_b$
	[11]	$L_{\tilde{f}} h(t,x) = \rho(t,x)\theta$, $\|\theta\| \leq \overline{\theta}$ $\hat{\theta}$ estimates θ and $\tilde{\theta}_U$ is upper error bound.	$\min\left\{\|\rho(t,x)\|\overline{\theta}, -\rho(t,x)\hat{\theta}(t) + \|\rho(t,x)\|\tilde{\theta}_U(t)\right\}$
	[14]	$\tilde{f}$ and $\tilde{g}$ are Lipschitz in x (with $L_{\tilde{f}}, L_{\tilde{g}}$), $\exists N$ data x_i, u_i and $\dot{x}_i$ so $\tilde{F}_i = \dot{x}_i - f(x_i) + g(x_i)u_i$	$\sum_{i=1}^{N} \left(\lambda_i^T \tilde{F}_i - \|\lambda_i\|(L_{\tilde{f}} + L_{\tilde{g}}\|u_i\|)\|x - x_i\|\right)$, λ_i are Lagrange multipliers.
	[16]	$\dot{h}_n(t,x,u) = 0$, $\dot{h}$ is Lipschitz in x and u (with L_x, L_u), $\exists N$ data x_i, u_i and $\dot{h}_i$	$\min_{i\in[1\cdots N]} \left[-\dot{h}_i + L_x\|x - x_i\| + L_u\|u - u_i\|\right]$
ISSf	[25]	Bounded uncertainty, continuously differentiable $\epsilon(r) > 0$ with $\frac{\mathrm{d}\epsilon}{\mathrm{d}r}(r) \geq 0$	$\frac{\|\nabla h(x)\|^2}{\epsilon(h(x))}$

A. Alan, et al., "*Parameterized Barrier Functions to Guarantee Safety Under Uncertainty*." IEEE CSL, 2023.



Robust safety filter

$$k(x) = \underset{u \in \mathcal{U}}{\operatorname{argmin}} \ \|u - k_{\text{nom}}(x)\|^2$$

$$\text{s.t. } \nabla h(x)^T f(x,u) - \sigma(x,u) \geq -\alpha(h(x))$$

Old habits die hard

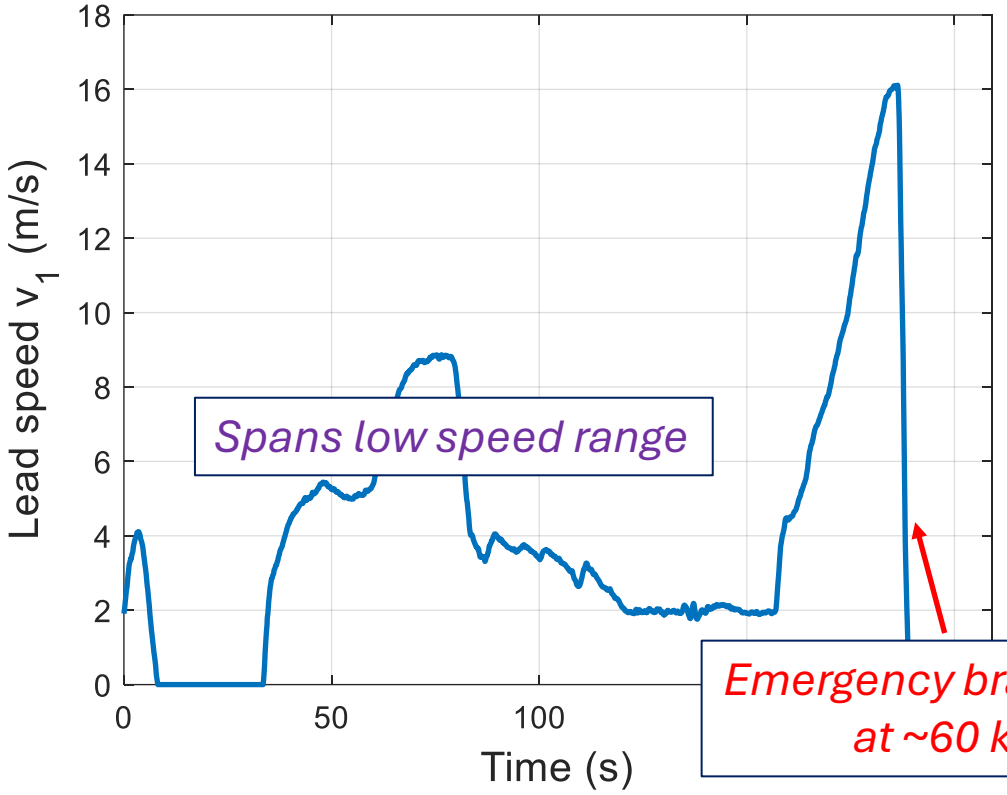


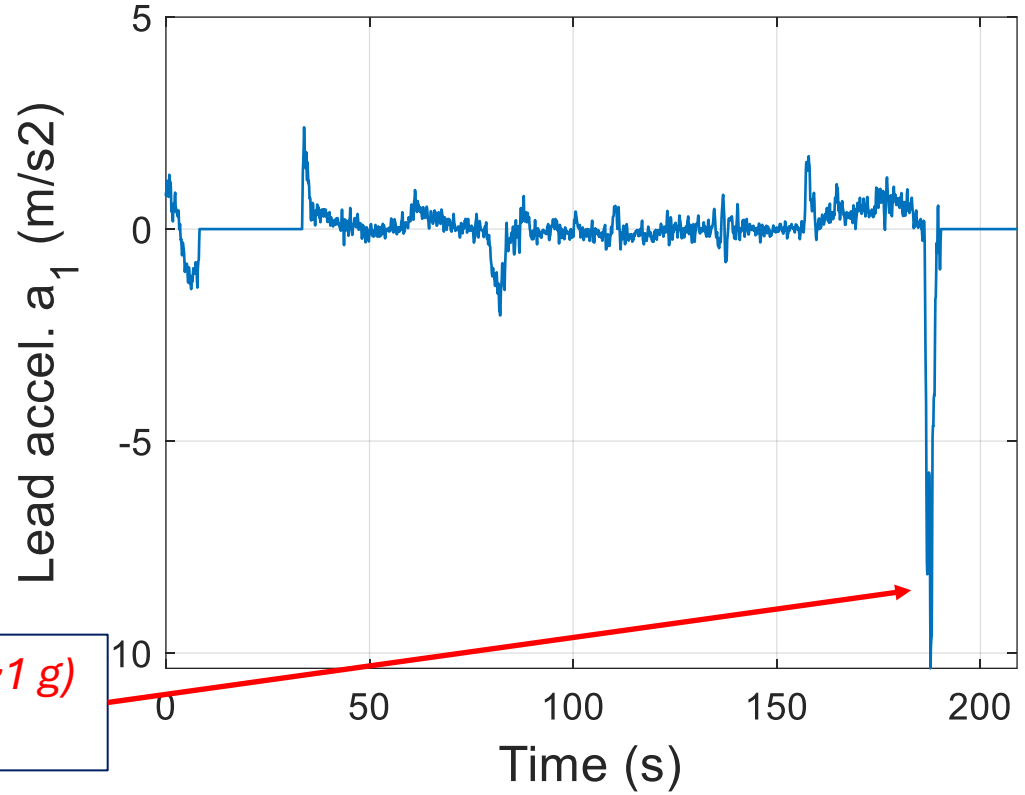


Thank you for your listening. *Questions?*

Preceding vehicle speed profile

Collected data from real traffic experiments:







Example: 1-D system

System: $\dot{x} = -x + \dfrac{1}{\arctan(x) + \pi} - d(t), \qquad \|d\|_\infty \leq 1$

Set: $S = \{x \in \mathbb{R} \mid h(x) = x \geq 0\}$

Ideal case ($d(t) = 0$): $\dot{h}(x) \geq -h(x), \ \forall x \geq 0 \implies$ safe.

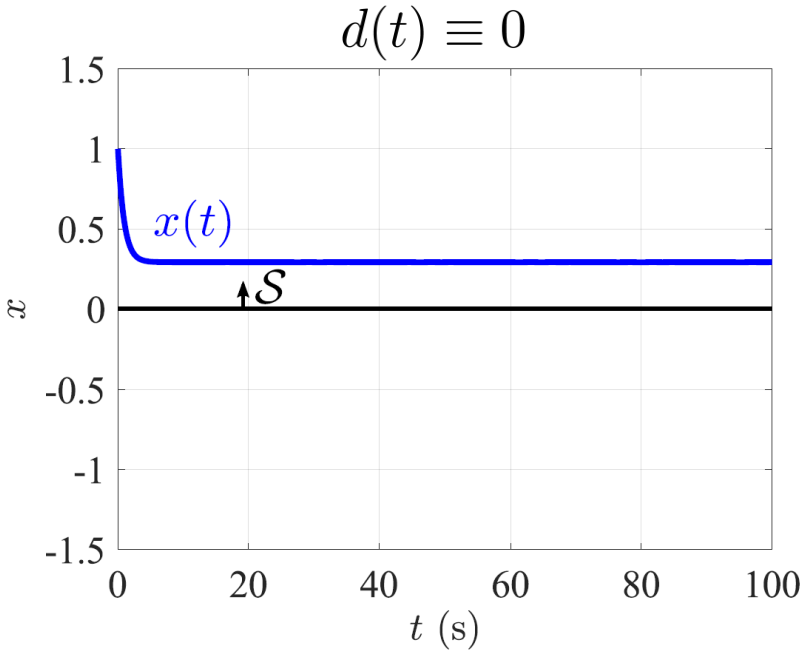
Robust case: *Not tunable*

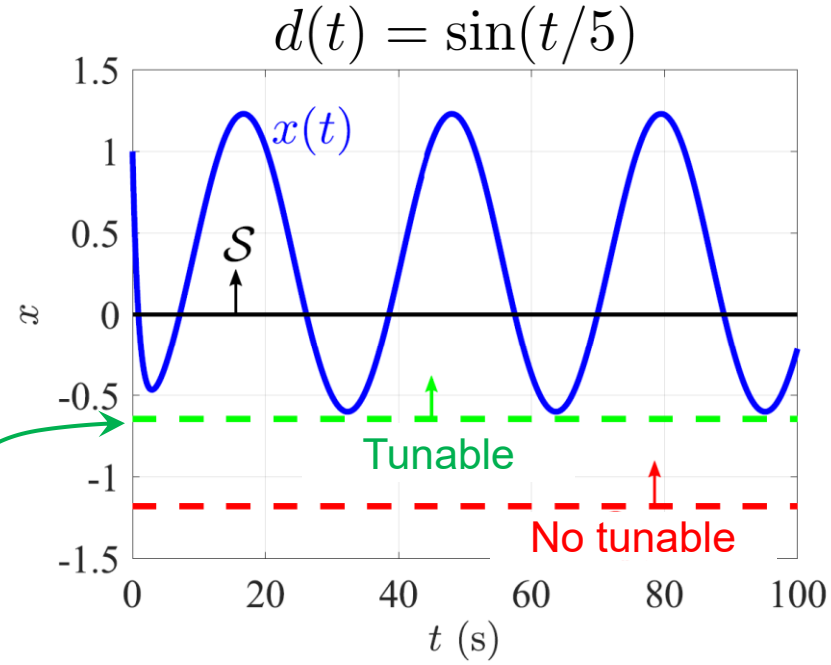
$\dot{h}(x, d) \geq -h(x) - \frac{3\pi\|d\|^2}{8} \implies$ ISSf with $\sigma(r) = \frac{3\pi r^2}{8} \in \mathcal{K}$.

Robust case: *Tunable*

$\dot{h}(x, d) \geq -h(x) - \frac{\|d\|^2}{4}(\arctan(x) + \pi) \implies$ ISSf.

Solved implicitly for
$\epsilon(r) = \arctan(r) + \pi$





Improved Sufficiency

Problem: $\dot{h}(x) \geq -\alpha(h(x)) + \|L_g h(x)\|^2 \epsilon$ does *not scale well* with disturbance.

Question: Can we find a more ***tuned*** modification that scales better?

Key idea: $\epsilon \to \epsilon(h(x))$ Example: $\epsilon(r) = \epsilon_0 \mathrm{e}^{-\lambda r},\ \epsilon_0 > 0,\ \lambda \geq 0$

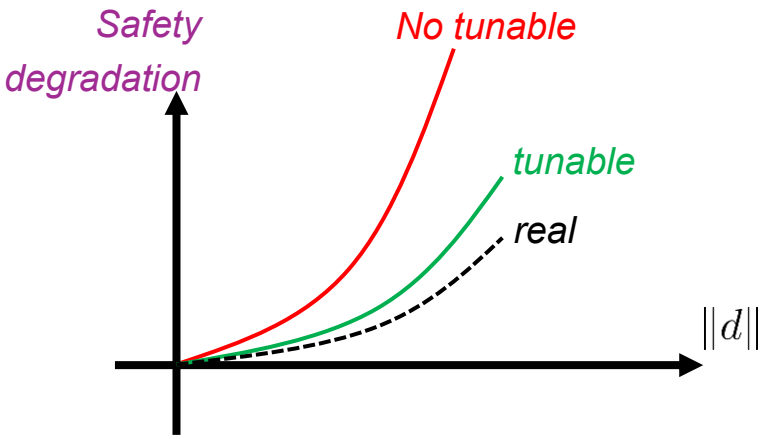
Theorem: Sufficiency with tunable ϵ

Let a function $\epsilon \in C^1(\mathbb{R}, \mathbb{R}_{>0})$ with $\epsilon'(r) \leq 0\ \forall r \in \mathbb{R}$.
If such a ϵ and $\alpha \in \mathcal{K}$ exist satisfying $\forall x \in \mathbb{R}^n$:

$$\dot{h}(x) \geq -\alpha(h(x)) + \|L_g h(x)\|^2 \epsilon(h(x)),$$

then system is ISSf Type-B with $\sigma_{\mathrm{T}}(\|d\|)$.
In particular, $\sigma_{\mathrm{T}}(\|d\|)$ is solution of the implicit function:

$$\sigma_{\mathrm{T}}(\|d\|) + \alpha^{-1}\left(-\frac{\|d\|^2}{4\epsilon(-\sigma_{\mathrm{T}}(\|d\|))}\right) = 0$$



Corollary: *Improvement*

Implicit Function Theorem for monotone equations:

- Such σ_{T} exists, is unique, and $\sigma_{\mathrm{T}} \in \mathcal{K}$.
- $\sigma_{\mathrm{T}}(\|d\|) \leq \sigma(\|d\|),\ \forall d \in \mathcal{D}$

A. Alan, et al., "*Safe controller synthesis with Tunable ISSf-CBFs*", IEEE CSL, 6, pp.908-913, 2022
A. Alan, et al., "*Control barrier functions and input-to-state safety with application to automated vehicles*", IEEE TCST, 31-6, 2023.